

# PRECISION AND RELIABILITY COSTS IN CONTRACTING CONTROL SYSTEMS

AUTHOR INFORMATION PENDING

**ABSTRACT.** We study the number of open-loop controls needed to keep actual initial states within a shrinking Euclidean neighbourhood of a state constraint. For binary planar systems contracting toward the origin, matched first derivatives determine an exact precision spectrum. We prove that, when neither controlled map has a critical point on the constraint, requiring failure probability  $\exp(-\alpha n + o(n))$  at precision  $\exp(-\gamma n + o(n))$  gives the rate  $\min\{S(\gamma), L_a(\alpha)\}$ . Here  $S$  is the full-state precision spectrum and  $L_a$  is the binary source-coding rate for a prescribed reliability exponent. Global injectivity and radius-independent dynamics are not required. The proof combines transport along all feasible transient branches with positive-area state regions that force prescribed prefixes for every approximately feasible input. An injective  $C^2$  example with null critical sets preserves every finite-horizon full-state catalogue count but increases the reliability cost. Thus noncollapse sufficient for ordinary precision costs need not preserve exponential reliability. We also determine the cost of one fixed nearly full-area compact set and isolate the role of first-order matching in the common low-precision rate.

## 1. INTRODUCTION

A controller can use knowledge of the initial state to select one input from a finite catalogue. The size of that catalogue measures the information needed to maintain a state constraint over a given time interval. Invariance entropy studies the exponential growth of such families; see [4, 7]. For systems that contract toward a common state under every input, a fixed positive environmental tolerance eventually makes the constraint automatic. The information cost can nevertheless remain positive when the tolerance decreases with the horizon. A further question arises when the catalogue is allowed to fail on a small set of initial states: how does the cost change if the permitted failure probability also decreases exponentially?

This paper considers these two scales together. The initial state lies in the triangle  $Q = \{(s, z) : 0 \leq s \leq 1, |z| \leq s\}$ , with its uniform area probability. An input succeeds when its trajectory lies within Euclidean distance  $\delta_n$  of  $Q$  at every time from 0 to  $n$ . The trajectory may leave and reenter  $Q$ . The chosen input may depend on the actual initial state, and every binary input is available. We let the tolerance and allowed failure satisfy

$$\delta_n = \exp(-\gamma n + o(n)), \quad e_n = \exp(-\alpha n + o(n)), \quad \gamma \geq 0, \quad \alpha > 0.$$

The successful set may change with  $n$ . This differs from fixing a compact initial set once and then covering it at every horizon.

Our systems are globally  $C^2$ , are controlled invariant on  $Q$ , and contract toward the origin in a common norm. Their first derivatives have two complementary angular branches of lengths  $a_0, a_1$ , with  $a_0 + a_1 = 1$ , and radial factors  $a_i^\beta$ ,  $\beta > 1$ . The actual dynamics can have angular feedback, truncated feasible images, and small overlapping feasible branches near the origin. They need not have a global inverse or a common exact radial formula. All structural assumptions are stated in Section 2.

---

2020 *Mathematics Subject Classification.* Primary 93C55, 37B40; Secondary 93D15, 94A24.

*Key words and phrases.* Control catalogues, invariance entropy, precision spectrum, reliability exponent, noncollapse, initial-state probability.

**1.1. Main answer and its mechanism.** Theorem 2.1 gives an exact joint cost when the derivatives are nonsingular on  $Q$ :

$$\lim_{n \rightarrow \infty} n^{-1} \log r_{e_n}(n, \delta_n; Q) = \min\{S(\gamma), L_a(\alpha)\}.$$

The two functions are explicit. The full-state spectrum  $S$  is determined by the angular entropy and radial stopping cost, whereas  $L_a$  is the classical binary source-coding rate at reliability exponent  $\alpha$ . The formula describes an initial-state index selecting an entire open-loop control. It is not a causal feedback or channel-capacity formula.

The distinction between ordinary cost and reliability is essential. Under the weaker assumption that each critical set has area zero, the full-state precision spectrum is still  $S$ . Every positive-area compact initial set has the same rate below the threshold  $\beta h$ , where  $h = -\sum_i a_i \log a_i$ . One fixed compact set of area arbitrarily close to one has rate  $\min\{h, \gamma/\beta\}$  simultaneously for all finite precision exponents. Above  $\beta h$  its cost is strictly less than that of the whole triangle (Theorem 2.3). However, null critical sets do not suffice for the joint reliability formula. Theorem 2.2 constructs an injective  $C^2$  system with no positive-area collapse whose full-state catalogue counts agree exactly with those of its linear model at every finite horizon and tolerance, while its reliability cost is strictly larger for a specified pair of exponents. A nonlinear transient changes the probabilities of difficult states without changing those full-cover counts.

The main difficulty is to relate symbolic estimates to regions covered by arbitrary approximate controls. Complete feasible regions can overlap, and a fixed-radius section may be empty or have a truncated terminal image. An input need not follow a designated feasible coding. Comparing it with such a coding at the first different symbol would therefore give an invalid lower bound. We instead use the first time it actually violates the state constraint. Theorem 2.5 bounds the physical area that one arbitrary input can cover, using separate angular and radial scales. This yields fixed-area precision estimates even through a finite noninjective transient.

Exponential reliability requires more. A discarded area of order  $\delta_n$  can be much larger than  $e_n$ , so the fixed-area estimate cannot simply be reused with a smaller exceptional set. We prove probability domination along every feasible transient branch, with one constant independent of the horizon and both budgets (Lemma 7.1). For the lower bound, Lemma 7.2 constructs disjoint positive-area regions of known mass on which every successful approximate input must follow a prescribed prefix. The first wrong input has a physical boundary excess larger than the tolerance, regardless of later reentry. These two estimates connect physical probability to classical type counting without assuming optimality of a coding or of a stopping tree.

**1.2. Relation to earlier work.** The finite-horizon full-cover quantity is outer invariance complexity in the sense of Chen and Zhong [2, Section 2.2]; their general framework already permits noninjective continuous maps. The invariance-entropy and control-set literature provides the underlying control-catalogue framework [4, 7]. Our question concerns simultaneous exponential precision and failure scales for a fixed physical initial-state distribution.

Measure-theoretic control entropy and transport of initial measures are established subjects. Wang, Huang and Sun [11] define measure invariance entropy using invariant partitions and their cylinder probabilities. Their Theorem 4.8 transports the initial measure together with a control conjugacy. Colonius [3] relates quasi-stationary metric invariance entropy to coder-controllers. Wang and Huang [10] study spanning-based measure entropy and inverse variational principles. Thus the use of probability-dependent catalogues is not itself a new contribution.

Nie, Wang and Huang [9] obtain measure entropy through convex duality from invariance pressure. Chen, Huang and Zhong [1] prove variational principles for entropy dimension using invariant-partition cylinders and time-power normalisation. Cost-threshold pressure formulas are developed by Yang, Chen, Yang and Zhou [12]. To use partition or pressure formulas for the quantity studied here, one must still identify the physical sets covered by arbitrary approximate inputs and control their mass with both budgets varying. A change of measure under conjugacy does not by itself compare two systems endowed with uniform physical area. The estimates in Lemmas 7.1 and 7.2 supply these quantitative links under our structural assumptions. The counterexample shows why they cannot be replaced by a qualitative null-set preservation statement.

Graph estimates, summable distortion, stopping scales, and typical-set arguments are classical tools; compare [6, 8]. The entropy and reliability optimisations below use elementary type counting, a standard source-coding method [5]. The contribution is the physical coverage and probability comparison that makes these tools applicable to unrestricted approximate control catalogues, together with the separation between ordinary precision and reliability under a null-critical transient. The explicit spectrum is a consequence of this geometry and the counting argument.

**1.3. Proof organisation.** Section 2 defines the objects and states the results. Sections 3–5 establish complete feasible-region bounds, finite-transient area control, and physical trajectories with a wrong-control margin. Section 6 derives the ordinary precision spectrum; Section 7 proves the reliability formula. The transient counterexample and a nonlinear example with inward boundary images follow in Sections 8 and 9. Appendix A uses the same two-scale geometry to characterise first-order matching within the specified matrix family. Appendix B records a finite-horizon caution about enlarging a control alphabet.

## 2. CONTROL CATALOGUES AND MAIN RESULTS

**2.1. Catalogue and precision.** Let

$$Q = \{(s, z) \in \mathbb{R}^2 : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{area}(Q) = 1.$$

For an infinite binary input  $u = (u_1, u_2, \dots)$  set  $F_u^0 = \text{id}$  and  $F_u^k = F_{u_k} \circ \dots \circ F_{u_1}$ . All binary inputs are allowed. Distances are Euclidean, and all logarithms are natural. For a nonempty compact  $K \subset Q$  define

$$(2.1) \quad A_u(n, \delta; K) = \{x \in K : \text{dist}(F_u^k x, Q) < \delta \text{ for every } 0 \leq k \leq n\}.$$

Let  $r_F(n, \delta; K)$  be the smallest number of infinite inputs whose sets (2.1) cover  $K$ . The assignment may depend on the actual initial state. Exit and reentry are allowed subject to every displayed constraint. Only the first  $n$  letters affect this count. Controlled feasibility makes it finite: all binary words of length  $n$ , with arbitrary infinite continuations, already provide a catalogue.

A precision sequence means  $\delta_n > 0$ ,  $\delta_n \rightarrow 0$  and  $-\log \delta_n/n \rightarrow \gamma$  for a finite  $\gamma \geq 0$ . It need not be monotone.

Let  $\mu = \text{area}|_Q$ , which is a probability since  $\text{area}(Q) = 1$ . For  $0 < e < 1$ , define

$$(2.2) \quad r_e(n, \delta; Q) = \min \left\{ \#\mathcal{U} : \mu \left( \bigcup_{u \in \mathcal{U}} A_u(n, \delta; Q) \right) \geq 1 - e \right\}.$$

The minimum is over finite catalogues of unrestricted binary inputs. It exists because the full-cover count is finite. A failure sequence has  $0 < e_n < 1$ ,  $e_n \rightarrow 0$  and  $-\log e_n/n \rightarrow \alpha > 0$ . Both sequences may be nonmonotone. The successful physical subset in (2.2) may change with  $n$ ; it is not a fixed compact initial set.

## 2.2. First-order data and global assumptions. Fix

$$(2.3) \quad a_0, a_1 \in (0, 1), \quad a_0 + a_1 = 1, \quad a_0 \neq a_1, \quad \beta_0, \beta_1 > 1,$$

and set  $r_i = a_i^{\beta_i}$ ,  $t_i = r_i/a_i < 1$ ,  $a_* = \min_i a_i$ ,  $a^* = \max_i a_i$  and  $h = -\sum_i a_i \log a_i$ . Write

$$(2.4) \quad A_0 = \begin{pmatrix} r_0 & 0 \\ t_0 a_1 & t_0 \end{pmatrix}, \quad A_1 = \begin{pmatrix} r_1 & 0 \\ -t_1 a_0 & t_1 \end{pmatrix}.$$

We retain separate radial exponents for the geometric estimates and the comparison in Appendix A. The reliability theorem uses their matched subfamily. The standing assumptions on the actual maps are:

- (i)  $F_0, F_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  are  $C^2$  maps,  $F_i(0) = 0$  and  $DF_i(0) = A_i$ .
- (ii)  $Q \subset F_0^{-1}Q \cup F_1^{-1}Q$ .
- (iii) For some  $\Lambda \geq 1$  and  $0 < \theta < 1$ , the norm  $V(s, z) = \max\{\Lambda|s|, |z|\}$  satisfies

$$(2.5) \quad V(F_i x) \leq \theta V(x) \quad (x \in \mathbb{R}^2, i = 0, 1).$$

- (iv) Each critical set

$$(2.6) \quad \mathcal{C}_i = \{x \in Q : \det DF_i(x) = 0\}$$

has area zero.

The second assumption is one-step controlled invariance; unrestricted concatenation yields an exactly feasible infinite input from each point of  $Q$ . The third concerns contraction toward the origin, not distances between two arbitrary trajectories. In particular it controls appended tails even outside  $Q$ . The fourth supplies noncollapse along feasible finite transients; it allows folds and requires no condition on critical points outside  $Q$ . Radial motion may depend on angle. No accumulated distortion, word-area bound or catalogue comparison is assumed.

For comparison, a fixed tolerance has zero exponential cost. Indeed  $\|x\|_2 \leq \sqrt{2}V(x)$ , and (2.5) gives a fixed  $J_\delta$  such that every input sends  $Q$  into the  $\delta$ -ball about zero after  $J_\delta$  steps and keeps it there. Choose an exactly feasible prefix of that length for each initial state and then append an arbitrary tail. The at most  $2^{J_\delta}$  prefixes cover  $Q$  for every longer horizon. Shrinking precision is what prevents this finite-transient argument from making the asymptotic question trivial.

Both matched and mismatched subfamilies are nonempty. For the linear maps  $A_i$ , the angular domains  $[-1, a_0 - a_1]$  and  $[a_0 - a_1, 1]$  cover every fibre. Put  $t = \max_i t_i < 1$  and choose  $\Lambda = 1 + 4t/(1 - t)$ . Then  $V(A_i x) \leq \max\{\max_i r_i, t(1 + 1/\Lambda)\}V(x)$  globally, with factor less than one, and  $\det A_i = r_i^2/a_i > 0$ .

For words write

$$\lambda_w = \prod_{j=1}^{|w|} a_{w_j}, \quad P_w = \prod_{j=1}^{|w|} r_{w_j},$$

with empty products one. Define

$$(2.7) \quad \chi_R = -\sum_i a_i \log r_i > h, \quad \gamma_0 = -\log \max_i r_i > 0, \quad r_0^D + r_1^D = 1.$$

The root  $D$  is unique in  $(0, 1)$ , since the left side decreases from two to zero and  $r_0 + r_1 < 1$ . Also  $\chi_R \geq \gamma_0$ .

**2.3. Probability costs.** Write  $H_b$  for binary entropy. In the matched case  $\beta_0 = \beta_1 = \beta$ , the angular cost and spectrum are

$$(2.8) \quad H_b(p) = -p \log p - (1 - p) \log(1 - p), \quad \chi(p) = -p \log a_0 - (1 - p) \log a_1,$$

$$(2.9) \quad S(\gamma) = \max_{p \in [0, 1]} H_b(p) \min\{1, \gamma/[\beta \chi(p)]\},$$

where  $0 \log 0 = 0$ . In particular  $H_b(a_0) = \chi(a_0) = h$ .

$$(2.10) \quad I_a(p) = \chi(p) - H_b(p), \quad L_a(\alpha) = \max_{I_a(p) \leq \alpha} H_b(p).$$

The function  $I_a$  is relative entropy with respect to  $(a_0, a_1)$ ;  $L_a$  is the classical binary source-coding rate at reliability exponent  $\alpha$ .

**Theorem 2.1** (Precision–reliability transfer). *Assume the standing hypotheses with  $\beta_0 = \beta_1 = \beta$ , and strengthen the critical-set condition to*

$$(2.11) \quad \det DF_i(x) \neq 0 \quad (x \in Q, i = 0, 1).$$

*For every fixed such system, every finite  $\gamma \geq 0$ , every  $\alpha > 0$ , and every precision and failure sequence of those exponents,*

$$(2.12) \quad \lim_{n \rightarrow \infty} \frac{\log r_{e_n}(n, \delta_n; Q)}{n} = \min\{S(\gamma), L_a(\alpha)\}.$$

*No global injectivity or global inverse is required.*

For a finite catalogue, enumerate its inputs and assign a successful state to its first successful input. The sets are Borel, so this is a measurable initial-state encoder. A fixed-length index sent once at time zero has minimum bit length  $\lceil \log_2 r_e \rceil$ ; the asymptotic bit rate is (2.12) divided by  $\log 2$ . There are no later messages. This interpretation does not identify the catalogue rate with a causal feedback data-rate or a channel capacity. The bit conversion is a definition-level consequence.

**Theorem 2.2** (A null-critical transient changes reliability). *For  $a_0 = 2^{-10}$ ,  $a_1 = 1 - 2^{-10}$  and  $\beta = 4096$ , there is a fixed injective  $C^2$  pair satisfying the standing matched hypotheses, with critical sets of area zero, such that*

$$(2.13) \quad r_F(n, \delta; Q) = r_A(n, \delta; Q) \quad (n \geq 1, \delta > 0).$$

*It maps no positive-area subset of  $Q$  into an area-zero set. Nevertheless, for  $\gamma = -\beta \log(a_0 a_1)$  and  $\alpha = 1/8$ , every pair of sequences with these exponents satisfies*

$$(2.14) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \frac{1}{2} \log 2 > \min\{S(\gamma), L_a(\alpha)\}.$$

Condition (2.11) is a sufficient condition for (2.12). The counterexample rules out replacing it merely by critical sets of area zero in a universal transfer theorem. It does not show that every system with critical points fails, or give a necessary and sufficient probability criterion.

**2.4. Ordinary precision costs and fixed initial sets.** The following results use only the standing null-critical hypothesis. They both describe the baseline precision cost and distinguish it from the exponentially reliable cost above.

**Theorem 2.3** (Precision spectrum and initial-set dependence). *Suppose  $\beta_0 = \beta_1 = \beta$  in the standing assumptions. For every fixed system, every finite  $\gamma \geq 0$  and every precision sequence of exponent  $\gamma$ ,*

$$(2.15) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = S(\gamma).$$

*Every fixed positive-area compact  $K \subset Q$  satisfies*

$$(2.16) \quad \min\{h, \gamma/\beta\} \leq \liminf_n \frac{\log r_F(n, \delta_n; K)}{n} \leq \limsup_n \frac{\log r_F(n, \delta_n; K)}{n} \leq S(\gamma).$$

*Thus the rate exists and is  $\gamma/\beta$  for  $0 \leq \gamma \leq \beta h$ . For each  $\zeta \in (0, 1)$  there is one fixed compact  $K_\zeta \subset Q$  with  $\text{area}(K_\zeta) > 1 - \zeta$  such that, simultaneously for every finite  $\gamma \geq 0$  and every precision sequence of that exponent,*

$$(2.17) \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \min\{h, \gamma/\beta\}.$$

The set is chosen before the exponent, sequence and horizon.

**Corollary 2.4** (Closed spectrum and two thresholds). *Under the hypotheses of Theorem 2.3, put  $a = a_* \in (0, 1/2)$ ,  $c_b = -\log(1 - a)$  and  $d_b = \log((1 - a)/a) > 0$ . The common-cost boundary and saturation boundary are respectively*

$$(2.18) \quad \gamma_* = \beta h, \quad \gamma_{\text{sat}} = -\frac{\beta}{2} \log(a_0 a_1) = \beta(c_b + d_b/2) > \gamma_*.$$

The full-set spectrum is

$$(2.19) \quad S(\gamma) = \begin{cases} \gamma/\beta, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b), & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

The exact gap between  $Q$  and the single set  $K_\zeta$  in the theorem is

$$(2.20) \quad \Delta(\gamma) = S(\gamma) - \min\{h, \gamma/\beta\} = \begin{cases} 0, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b) - h, & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2 - h, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

It is strictly positive exactly when  $\gamma > \gamma_*$ . The full-set rate equals  $\log 2$  exactly when  $\gamma \geq \gamma_{\text{sat}}$ .

The first threshold concerns equality of rates across all positive-area compact initial sets; the second concerns the full-set catalogue alone. The theorem does not classify high-side rates of arbitrary compact initial sets or assert that only two such rates are possible.

The role of matching can also be tested within the same matrix family. Appendix A proves that unequal radial powers give distinct full-state and near-full-set rates already for every  $0 < \gamma < \gamma_0$ . Thus a common positive-area rate on an interval adjacent to zero holds precisely when the powers match. This criterion concerns ordinary precision costs in the specified family; it is not a classification of reliability.

**2.5. The physical coverage estimate.** The following estimate bounds the mass covered by one arbitrary input, without selecting an exact coding. It supplies the lower rate for positive-area initial sets in both cases. Its regular compact set also provides the single near-full set used in the upper bounds. These are fixed-loss estimates: their  $C_T \delta$  term is not assigned to an exponentially small probability budget.

**Theorem 2.5** (Area covered by an arbitrary approximate input). *Under the standing assumptions, whether matched or mismatched, there is a fixed integer  $J \geq 1$  with the following property. For every positive-area compact  $K \subset Q$  and every  $0 < \varepsilon < \text{area}(K)$ , one can choose a compact  $T \subset K \setminus \{0\}$  with  $\text{area}(T) > \text{area}(K) - \varepsilon$  and  $C_T < \infty$  so that for every  $e \in (0, h)$  there is  $C_{T,e} < \infty$  such that*

$$(2.21) \quad \text{area}(T \cap A_u(n, \delta; K)) \leq C_T \delta + C_{T,e} \left[ e^{-(h-e)m} + \delta e^{(\chi_R - h + 2e)m} \right]$$

for every input  $u$ , integers  $n \geq J$  and  $0 \leq m \leq n - J$ , and  $0 < \delta \leq 1$ . The integer depends only on the fixed system. The set and  $C_T$  precede  $e, n, \delta, u$ ; the second constant may also depend on  $e$ . In particular, for  $K = Q$  the set can have area greater than  $1 - \zeta$  for any prescribed  $\zeta \in (0, 1)$ .

### 3. LOCAL FEASIBLE GEOMETRY

We first control complete feasible regions near the origin. The angular and radial products describe different scales, so they remain separate throughout this section. The first-exit estimate will then apply to any input, even one that is only approximately feasible.

Use  $(s, y)$  with  $z = sy$ , only for  $s > 0$ . Define

$$R_i(s, y) = F_i^s(s, sy)/s, \quad N_i(s, y) = F_i^z(s, sy)/s, \quad T_i = N_i/R_i.$$

The integral formula  $F_i(s, sy)/s = \int_0^1 DF_i(ts, tsy)(1, y) dt$  extends  $R_i, N_i$  as  $C^1$  functions to  $s = 0$ . At zero

$$(3.1) \quad R_i(0, y) = r_i, \quad L_0(y) := T_0(0, y) = (y + a_1)/a_0, \quad L_1(y) := T_1(0, y) = (y - a_0)/a_1.$$

We choose the local constants from the maps as follows. Let  $L$  bound the Euclidean bilinear operator norm of both second derivatives on  $\bar{B}(0, 2)$ , and let  $r_* = \min r_i$ ,  $r^* = \max r_i$ ,  $a^* = \max a_i$ ,  $a_* = \min a_i$ . Take  $B = 10^4(1 + L)/r_*^3$ , and shrink  $\rho > 0$  so that

$$(3.2) \quad \rho \leq \min \left\{ \frac{1}{4}, \frac{r_*}{8(1 + L)}, \frac{1 - r^*}{8B}, \frac{1/a^* - 1}{16B}, \frac{1}{16B}, \frac{a_*}{16B} \right\}.$$

Taylor's integral formula gives  $|R_i - r_i| \leq 2Ls$ ,  $|\partial_s R_i|, |\partial_s N_i| \leq 2L$ , and  $|\partial_y R_i| \leq 2Ls$ ,  $|\partial_y N_i - t_i| \leq 2Ls$ . The same bounds apply with the appropriate constant linear term in  $N_i$ . In particular  $R_i \geq r_*/2$ . Dividing these formulas by  $R_i$ , and differentiating the quotient, gives, with the generous common constant  $B$ ,

$$(3.3) \quad \begin{aligned} |R_i - r_i| &\leq Bs, & |\log(R_i/r_i)| &\leq Bs, \\ |\partial_s \log R_i| &\leq B, & |\partial_y \log R_i| &\leq Bs, \\ |T_i - L_i| &\leq Bs, & |\partial_s T_i| &\leq B, & |\partial_y T_i - 1/a_i| &\leq Bs. \end{aligned}$$

For example  $|N_i| \leq 3$  on this rectangle; quotient derivatives use  $R_i^{-2} \leq 4/r_*^2$ . Thus the displayed choice of  $B$  dominates all the constants in (3.3). These are derived local inequalities, not extra hypotheses.

Set  $\bar{r} = r^* + B\rho < 1$  and  $\hat{r} = r^* + 2B\rho < 1$ . On an exactly feasible local orbit,  $s_k \leq \bar{r}^k s_0$ . The bounds in (3.2) also give  $T_1(s, -1) < -1$  and  $T_0(s, 1) > 1$ . Controlled invariance therefore forces

$$(3.4) \quad T_0(s, -1) \geq -1, \quad T_1(s, 1) \leq 1.$$

Equality is not required in (3.4). The increasing functions  $T_i(s, \cdot)$  have unique cut points  $\vartheta_0(s)$  with  $T_0 = 1$  and  $\vartheta_1(s)$  with  $T_1 = -1$ . They equal  $a_0 - a_1 + O(s)$ . Their feasible intervals are respectively  $[-1, \vartheta_0(s)]$  and  $[\vartheta_1(s), 1]$ . Controlled invariance gives  $\vartheta_1 \leq \vartheta_0$ . Thus any local overlap has angular width  $O(s)$  and physical width  $O(s^2)$ . Positivity of the overlap is not assumed.

**3.1. Complete intervals with truncated images.** Consider a radial graph  $s = S(y)$ , on any closed subinterval of  $[-1, 1]$ , with  $|S'/S| \leq 1$ . Put  $k = S'/S$ . For either input its terminal angular derivative along the graph is

$$J_i = \partial_y T_i + \partial_s T_i S' = 1/a_i + O(BS) > 0.$$

The new radial graph has logarithmic slope

$$(3.5) \quad k_{\text{new}} = \frac{k + (\partial_s \log R_i)S' + \partial_y \log R_i}{J_i}.$$

Its numerator is at most  $1 + 2B\rho$  in absolute value and its denominator at least  $1/a^* - 2B\rho$ . By (3.2),  $|k_{\text{new}}| \leq 1$ . Piecewise  $C^1$  graphs obey the same assertion almost everywhere.

For a fixed initial radius  $0 < s_0 \leq \rho$ , start with the horizontal graph  $S \equiv s_0$ . Let  $I_w(s_0)$  be the set of initial angles whose trajectory under a finite word  $w$  stays in  $Q$  at every intermediate time. It is a closed interval, possibly empty or a point. Here is the induction that checks all constraints. The current angular image is an interval, with a graph obeying (3.5). Extend that graph constantly to the two sides of its domain. The extension is positive, bounded by  $\rho$ , and obeys the logarithmic-slope bound almost everywhere. Both  $T_i(S(y), y)$  are increasing. By (3.4), control 0 cannot violate the lower angular boundary, and control 1 cannot violate the upper one. The other boundary imposes one cut. Pulling

that cut back through the increasing preceding angular map leaves an interval. No terminal map is asserted to be onto  $[-1, 1]$ . The two allowed children cover the parent, by controlled invariance, but either child can be empty.

Along any such feasible graph, (3.3) and the geometric decay of  $s_j$  give constants independent of  $w, s_0$  and its initial angle:

$$(3.6) \quad C_R^{-1} s_0 P_w \leq s_{|w|} \leq C_R s_0 P_w, \quad C_A^{-1} \lambda_w^{-1} \leq \frac{dY_w}{dy_0} \leq C_A \lambda_w^{-1}.$$

For instance one may take  $C_R = \exp(B\rho/(1 - \bar{r}))$ ,  $C_A = \exp(4B\rho/(1 - \bar{r}))$ :  $|a_i J_i - 1| \leq 2Bs_j \leq 1/8$ , so its logarithm has modulus at most  $4Bs_j$ . The first estimate sums  $\log(R_i/r_i)$ ; the second sums the angular derivatives. They are independent product estimates. Only in the matched case is  $P_w = \lambda_w^\beta$ . Integration over the *actual* angular image implies

$$(3.7) \quad |I_w(s_0)| \leq 2C_A \lambda_w, \quad |\mathcal{I}_w| \leq C_A \rho^2 \lambda_w, \quad \mathcal{I}_w = \{(s, sy) : 0 < s \leq \rho, y \in I_w(s)\}.$$

The Jacobian of  $(s, y) \mapsto (s, sy)$  is  $s$ . There is no positive lower bound on  $|I_w|$ ; some fixed-radius slices in Section 9.1 are empty.

For a point interval the length bound is zero. Derivative comparisons on nondegenerate intervals extend to their endpoints by continuity; degenerate fibres remain in all covers and do not contribute area.

**3.2. The first true exit.** For a fixed input, partition its covered states by the first time at which its trajectory actually leaves  $Q$ . Up to that time the prefix is exactly feasible, so the preceding interval estimates apply. The permitted tolerance restricts the states that can survive this first exit.

Suppose the prefix  $w$  of length  $j - 1$  is exactly feasible and the next prescribed input  $i$  is the first one to leave  $Q$ . On the current graph the map  $T_i$  is increasing with derivative at least  $1/(2a_i)$ . The radial output is at least  $cs_0 P_w$ , by (3.3), (3.6), with  $c > 0$  fixed. If this wrong step nonetheless has distance less than  $\delta$  from  $Q$ , then

$$(3.8) \quad (|z_j| - s_j)_+ < 2\delta.$$

This follows from the Euclidean Lipschitz linear functionals  $z - s$  and  $-z - s$ . The output radius is positive and at most  $\rho$ , so the violated constraint is angular, not the upper radius. Consequently the wrong part of the terminal angular interval has diameter at most  $C\delta/(s_0 P_w)$ . By (3.6), its initial-angle preimage has length at most

$$C\delta \lambda_w / (s_0 P_w).$$

After integration against  $s_0 ds_0$ , its physical area is at most

$$(3.9) \quad C\delta \lambda_w / P_w.$$

This argument also covers a point interval or a terminal interval lying wholly beyond the cut: monotonicity bounds the diameter of the part satisfying (3.8), without requiring a cut point inside the actual image. It does not require the wrong trajectory to remain outside afterwards. Any later reentry cannot erase constraint (3.8) at its first exit. A different exact feasible coding is not counted as an exit and has no role in (3.9).

#### 4. FINITE TRANSIENTS AND ACTUAL COVERED AREA

Common contraction gives a uniform time of entry into the local region. To transport its area estimates back to the initial set, we remove a small area once and retain all feasible transient choices on the remaining compact window. This is where the null-critical-set hypothesis is used; a global inverse is not required.

For a finite word  $w$ , let

$$(4.1) \quad D_w = \{x \in Q : F_{w|k} x \in Q \ (1 \leq k \leq |w|)\}, \quad D_\emptyset = Q.$$

These sets are compact. Write  $w^-$  for the word with its last letter removed. At a first true exit under  $w$ , all derivative inputs still lie in  $Q$ , so the relevant domain is  $D_{w^-}$ . The last output need not belong to  $Q$ .

**Lemma 4.1** (Null pullbacks along feasible prefixes). *For every Lebesgue null set  $B \subset \mathbb{R}^2$ , every control  $i$ , and every finite word  $w$ ,*

$$(4.2) \quad \text{area}(Q \cap F_i^{-1}B) = 0, \quad \text{area}(D_w \cap F_w^{-1}B) = 0.$$

*For each nonempty word the compact set*

$$(4.3) \quad Z_w = D_{w^-} \cap \{x : \det DF_w(x) = 0\}$$

*also has area zero.*

*Proof.* The inverse function theorem supplies an open inverse chart at every point of  $Q \setminus \mathcal{C}_i$ . A countable family of these charts covers that set. Exhaust each image by relatively compact subcharts; its  $C^1$  inverse is Lipschitz on sufficiently small compact neighbourhoods. A Lipschitz map preserves null sets: cover a bounded null set by squares with arbitrarily small total area, and cover their images by discs whose total area is bounded by a constant times that sum. Thus every chart pullback of  $B$  is null. The omitted  $\mathcal{C}_i$  is null by (2.6). A Borel null superset handles an arbitrary Lebesgue null set.

For  $w = vi$ , exact feasibility gives

$$D_w \cap F_w^{-1}B \subset D_v \cap F_v^{-1}(Q \cap F_i^{-1}B).$$

Induction proves the second assertion of (4.2), including the empty word. Finally, on  $D_{w^-}$  the chain rule expresses  $\det DF_w$  as a product of determinants evaluated at states in  $Q$ . If the factor at time  $j$  vanishes, the initial state belongs to

$$D_{w|j-1} \cap F_{w|j-1}^{-1}\mathcal{C}_{w_j}.$$

This is null by (4.2). Their finite union contains  $Z_w$ , which is compact by continuity and closedness of the feasibility domain.  $\square$

Choose  $J \geq 1$ , depending only on the fixed system, so that

$$(4.4) \quad \theta^J \sup_Q V \leq \rho.$$

Every exactly feasible  $J$ -prefix ends in  $Q_\rho = Q \cap \{s \leq \rho\}$ .

**Lemma 4.2** (One regular compact window). *For every positive-area compact  $K \subset Q$  and  $0 < \varepsilon < \text{area}(K)$ , there is a compact  $T \subset K \setminus \{0\}$ , with  $\text{area}(T) > \text{area}(K) - \varepsilon$ , and  $L_T < \infty$  such that*

$$(4.5) \quad \text{area}(T \cap D_{w^-} \cap F_w^{-1}B) \leq L_T \text{area}(B)$$

*for all  $1 \leq |w| \leq J$  and all Lebesgue measurable  $B \subset \mathbb{R}^2$ . On the same set, for every  $e \in (0, h)$  there is  $D_e \geq 1$  such that every local exactly feasible tail  $v$ , after any exactly feasible  $J$ -prefix from  $T$ , satisfies*

$$(4.6) \quad \begin{aligned} D_e^{-1}e^{-(h+e)|v|} &\leq \lambda_v \leq D_e e^{-(h-e)|v|}, \\ D_e^{-1}e^{-(\chi_R+e)|v|} &\leq P_v \leq D_e e^{-(\chi_R-e)|v|}. \end{aligned}$$

*The set and  $L_T$  precede  $e$ , every input, tolerance and horizon.*

*Proof.* For  $d > 0$ , the total weight  $\sum \lambda_v$  of length- $m$  words with  $|\#0/m - a_0| > d$  is at most  $2e^{-2d^2m}$ . Indeed, for a Bernoulli variable  $X$  of mean  $a_0$ , the second derivative of  $\log \mathbb{E}e^{t(X-a_0)}$  is a tilted variance, at most  $1/4$ . Its value and derivative at zero vanish, so it is at most  $t^2/8$ . Exponential Markov with  $t = 4d$  and  $t = -4d$  gives the bound. By (3.7), the unions of the corresponding complete local feasible regions have summable area. Borel–Cantelli for rational  $d > 0$  gives a Borel null set  $B_{\text{loc}} \subset Q_\rho$ , including the origin,

outside which all feasible length- $m$  prefixes have zero frequency tending uniformly to  $a_0$ . Overlaps do not invalidate this union bound. Controlled invariance supplies at least one feasible prefix at every length.

Lemma 4.1 shows that

$$B_J = \bigcup_{|\sigma|=J} (D_\sigma \cap F_\sigma^{-1} B_{\text{loc}})$$

is null. The finite union  $Z_J = \bigcup_{1 \leq |w| \leq J} Z_w$  is a compact null set. On  $Q \setminus B_J$ , the maximum frequency deviation over all feasible transient choices and all their feasible length- $m$  tails tends to zero. It is measurable: for each  $m$  there are finitely many words and their complete feasibility domains are closed. Egorov on  $K$ , followed by removal of  $B_J \cup Z_J \cup \{0\}$  and compact inner approximation, gives  $T$  of the stated area with uniform convergence. Allocate less than  $\varepsilon/2$  loss to Egorov and less than  $\varepsilon/2$  to inner approximation. Both logarithmic products are affine functions of frequency. Uniform convergence gives (4.6); finitely many early lengths are absorbed in  $D_\varepsilon$ . This selects one set before all precision choices, rather than a set for each exponent.

For a word  $w$  with  $1 \leq |w| \leq J$ , the compact set  $M_w = T \cap D_{w-}$  is disjoint from  $Z_w$ . The composition  $F_w$  is therefore regular at every point of  $M_w$ . Cover it by finitely many open inverse charts  $U_{w,k}$ , with compact closures inside regular inverse charts. Change of variables on each chart gives

$$\text{area}(M_w \cap U_{w,k} \cap F_w^{-1} B) \leq \left( \sup_{\bar{U}_{w,k}} |\det DF_w|^{-1} \right) \text{area}(B).$$

The supremum is finite. Sum over the charts and take the maximum over the finite word family, omitting empty  $M_w$ 's, to obtain (4.5). Chart overlaps only enlarge the bound; every local inverse branch meeting  $M_w$  is included.  $\square$

The lemma provides an inverse-area bound on a fixed regular window, not a Jacobian lower bound on all  $Q$  or a global multiplicity bound. Its constants may depend on the prescribed area loss. The upper catalogues below still cover every actual state, including the critical states omitted from this window.

*Proof of Theorem 2.5.* Choose  $J, T, L_T$  from Lemma 4.2. Let  $Q_\delta = \{x : \text{dist}(x, Q) < \delta\}$ . For  $0 < \delta \leq 1$ , the tubes around the three edges of  $Q$  give

$$\text{area}(Q_\delta \setminus Q) \leq C_Q \delta, \quad C_Q = 2(2 + 2\sqrt{2}) + 3\pi.$$

If a state in  $T \cap A_u(n, \delta; K)$  first truly exits at  $j \leq J$ , put  $w = u|j$ . Its earlier states lie exactly in  $Q$ , and it belongs to

$$T \cap D_{w-} \cap F_w^{-1}(Q_\delta \setminus Q).$$

By (4.5), its class has area at most  $L_T C_Q \delta$ . Summing over the  $J$  possible first-exit times gives

$$(4.7) \quad \text{area}\{x \in T \cap A_u(n, \delta; K) : \text{first true exit occurs by } J\} \leq C_T \delta, \quad C_T = J L_T C_Q.$$

All derivative inputs here are in  $Q$ ; no nonsingularity condition is needed at the last output, which may be outside.

For the remaining states,  $\sigma = u|J$  is exactly feasible. Transport them by  $F_\sigma$  to  $Q_\rho$ , and use (4.5) to pull back endpoint area estimates. If the next  $m$  prescribed digits are exactly feasible for at least one transported state, (3.7) and (4.6) bound the area of the survival class by  $C_{T,e} e^{-(h-e)m}$ . If no such state exists, that class is empty. For a first true local exit at time  $j \leq m$ , the prescribed  $(j-1)$ -prefix is feasible for a transported state whenever the class is nonempty. Its pulled-back area, by (3.9), (4.5) and both products in (4.6), is at most

$$C_{T,e} \delta e^{(\chi_R - h + 2e)(j-1)}.$$

Since  $\chi_R > h$ , the sum is bounded by a fixed multiple of  $\delta e^{(\chi_R - h + 2e)m}$ . Together with (4.7) and exact survival, this proves (2.21). A switch between feasible codes was never counted as an exit. Every covered state satisfies the constraint at its first true exit, regardless of subsequent reentry.  $\square$

## 5. PHYSICAL TRAJECTORIES WITH A WRONG-CONTROL MARGIN

The area estimate concerns positive-area mass. For the full constraint we also need actual states whose controls cannot be replaced by approximate alternatives. We construct them at one fixed positive radius for each run bound, then force prefixes while their physical margin remains larger than the tolerance.

**Lemma 5.1** (Bounded-run realisation). *For each integer  $L \geq 2$ , there is a positive radius  $s_L$  and  $d'_L > 0$  such that, for every  $0 < s \leq s_L$ , every infinite binary sequence whose successive runs have length at most  $L$  is realised by an actual exactly feasible orbit starting at  $(s, sy_0)$ . At every step of that orbit the alternative control is genuinely infeasible, with an angular excess at least  $d'_L$ . The initial-angle centre can be chosen Lipschitz in  $s$ , with a constant uniform over the sequences. The radius bound and margin are uniform over the sequences. No such assertion is made for unbounded runs or all constant words.*

*Proof.* The inverse reference maps are  $f_0(y) = a_0y - a_1$ ,  $f_1(y) = a_1y + a_0$ . For any fixed infinite input  $v$ , successive inverse compositions give a unique reference bounded orbit  $y_k^0 \in [-1, 1]$ . If runs have length at most  $L$ , the next opposite digit is seen within  $L + 1$  digits. Since  $f_0(y) + 1 = a_0(y + 1)$ ,  $1 - f_1(y) = a_1(1 - y)$ , and each opposite digit inserts an endpoint distance at least  $2a_*$ ,

$$(5.1) \quad |y_k^0| \leq 1 - d_L, \quad d_L = 2a_*^{L+1} > 0.$$

Let  $b = d_L/4$  and consider the complete closed ball of sequences  $\|y - y^0\|_\infty \leq b$  in  $\ell^\infty$ . For each such sequence and a positive fixed initial radius  $s_*$ , recursively define  $s_{k+1} = s_k R_{v_{k+1}}(s_k, y_k)$ . These radii stay below  $\bar{r}^k s_*$ , independently of whether the angular recursion has yet been imposed. The scalar map  $s \mapsto s R_i(s, y)$  has derivative bounded in absolute value by  $\hat{r} < 1$ , and its  $y$  derivative is bounded by  $Bs^2$ , since  $R_i \leq \bar{r} < 1$  and (3.3) bounds its logarithmic angular derivative. Thus two candidate sequences give

$$(5.2) \quad \sup_k |s_k(y) - s_k(\tilde{y})| \leq C_s s_*^2 \|y - \tilde{y}\|_\infty, \quad C_s = B/(1 - \hat{r}).$$

Write  $e_i(s, y) = T_i(s, y) - L_i(y)$ . By (3.3),  $|e_i| \leq Bs$ ,  $|\partial_s e_i| \leq B$ , and  $|\partial_y e_i| \leq Bs$ . Define

$$(5.3) \quad (\mathcal{B}y)_k = f_{v_{k+1}}(y_{k+1}) - a_{v_{k+1}} e_{v_{k+1}}(s_k(y), y_k).$$

Choose  $s_* \leq \rho$  so small that

$$(5.4) \quad a^* B s_* \leq (1 - a^*)b/2, \quad a^* B(s_* + C_s s_*^2) < (1 - a^*)/2.$$

Then (5.3) maps the ball into itself and has Lipschitz constant at most  $a^* + a^* B(s_* + C_s s_*^2) < 1$ . The contraction mapping theorem gives a fixed sequence. Its equation is exactly  $y_{k+1} = T_{v_{k+1}}(s_k, y_k)$ , not a formal symbolic substitution. It has  $|y_k| \leq 1 - 3d_L/4$ , so the constructed physical trajectory is exactly feasible for all times.

To check the alternative control, suppose the chosen digit is zero. Then  $T_0(s_k, y_k) = y_{k+1} \leq 1 - 3d_L/4$ , and (3.3) gives

$$y_k \leq a_0 - a_1 - 3a_0 d_L/4 + a_0 B s_k.$$

Substitution into  $T_1 = L_1 + e_1$  yields

$$T_1(s_k, y_k) \leq -1 - \frac{3a_0 d_L}{4a_1} + B(1 + a_0/a_1)s_k.$$

For chosen digit one, the symmetric computation gives

$$T_0(s_k, y_k) \geq 1 + \frac{3a_1 d_L}{4a_0} - B(1 + a_1/a_0)s_k.$$

Shrink  $s_*$  once more, before considering any particular sequence, so that these last error terms are at most  $a_* d_L/(4a_*)$ . Both wrong angular excesses are then at least  $d'_L = a_* d_L/(2a_*)$ . Set  $s_L = s_*$ . All choices depend only on  $L$  and the fixed system. The same estimates hold for every initial radius  $0 < s \leq s_L$ . For a fixed candidate sequence, scalar contraction gives  $\sup_k |s_k(s, y) - s_k(s', y)| \leq |s - s'|$ . If  $\kappa_L < 1$  is the common contraction constant of (5.3), the fixed-point equations therefore give

$$(5.5) \quad \|y^v(s) - y^v(s')\|_\infty \leq \frac{a^* B}{1 - \kappa_L} |s - s'|.$$

This proves the uniform physical realisation, continuity, and genuine-infeasibility assertions.  $\square$

The preceding margin is angular. Its physical size decreases with the radius, so a long exact word need not remain distinguishable at an assigned tolerance. The next lemma chooses a shorter prefix and compares its radial scale directly with that tolerance.

**Lemma 5.2** (Forcing a block prefix). *Fix  $L \geq 2$  and  $1 \leq j \leq L - 1$ , and put*

$$M_L = \binom{L-2}{j-1}, \quad c_L = -\frac{j}{L} \log r_0 - \left(1 - \frac{j}{L}\right) \log r_1.$$

*For every precision sequence of exponent  $\gamma > 0$  and every  $0 < t < \min\{1, \gamma/c_L\}$ , for all sufficiently large  $n$ ,*

$$(5.6) \quad r_F(n, \delta_n; Q) \geq M_L^{\lfloor tn/L \rfloor}.$$

*Proof.* Use all length- $L$  blocks starting with zero, ending with one, and containing  $j$  zeros. Every concatenation has runs of length at most  $L$  and radial product  $e^{-Lc_L}$  per complete block. Put  $k_n = \lfloor tn/L \rfloor$  and  $m_n = Lk_n$ . For each of the  $M_L^{k_n}$  initial concatenations, append one fixed allowable block forever. Lemma 5.1 realises these infinite inputs at the same positive radius  $s_L$ . Every prefix through time  $m_n$  has  $P_{v|k} \geq P_{v|m_n} = e^{-c_L m_n}$ .

If an arbitrary catalogue input first differs from a witness at  $j' \leq m_n$ , its preceding states agree with that witness. By (3.6), their radius at  $j' - 1$  is at least  $C_R^{-1} s_L e^{-c_L m_n}$ . The wrong radial factor is at least  $r_*/2$ , and the alternative angular excess is at least  $d'_L$ . Its actual physical excess therefore obeys

$$(5.7) \quad |z_{j'}| - s_{j'} \geq E_L e^{-c_L m_n}, \quad E_L = \frac{s_L r_* d'_L}{2C_R} > 0.$$

The ratio to  $2\delta_n$  has logarithmic rate  $\gamma - c_L t > 0$ . For large  $n$ , it violates (3.8), regardless of later reentry. Thus the approximate input must agree with all  $m_n$  witness digits. Distinct selected prefixes give distinct actual states: at their first different digit one input is feasible and the other has the proved wrong-control margin. One catalogue input covers at most one of them, proving (5.6). The block length, positive radius, and margin remain fixed throughout this horizon limit. Integer rounding and every  $o(n)$  precision fluctuation are absorbed by the strict choice of  $t$ .  $\square$

## 6. STOPPING CATALOGUES AND THE MATCHED SPECTRUM

We first construct catalogues by following feasible inputs until the radius is small enough for a common tail. This construction applies under all the standing assumptions. We then prove the matched spectrum using the physical prefix lower bound and the covered-area estimate. The reference tree provides an upper bound; its leaves need not all be realised by actual states.

**Lemma 6.1** (A complete radial stopping catalogue). *Put  $M = \max\{1, 4\sqrt{2}\Lambda C_R\rho\}$ . For  $N \geq 0$  and  $0 < t \leq 1$ , let  $\mathcal{L}_P(N, t)$  be the full reference binary tree stopped at the first  $P_w \leq t$ , or at depth  $N$ , whichever occurs first. For  $n \geq J$  and  $0 < \delta \leq 1$ ,*

$$(6.1) \quad r_F(n, \delta; Q) \leq 2^J \# \mathcal{L}_P(n - J, \delta/M).$$

*The catalogue can retain only transient/leaf pairs whose complete actual feasible initial region is nonempty.*

*Proof.* For every actual initial state choose an exactly feasible infinite input. Its first  $J$  letters form a feasible transient ending in  $Q_\rho$ ; the subsequent letters reach a reference leaf. Keep every pair realised by an actual state, including singleton regions and boundary states. Before stopping the assigned states are exactly feasible. At a threshold leaf, (3.6) gives

$$s_{|w|} \leq C_R \rho P_w \leq C_R \rho \delta / M \leq \delta / (4\sqrt{2}\Lambda).$$

Since  $V = \Lambda s$  on  $Q$ , appending the same infinite zero tail and using (2.5) keeps the Euclidean norm at most  $\delta/4$  at every later time. The origin belongs to  $Q$ , so every constraint holds strictly on that tail. A horizon leaf is exactly feasible through time  $n$ , and its later extension is irrelevant. Every constructed input fixes the origin as well. There are at most  $2^J$  transients, proving (6.1).  $\square$

**6.1. Upper bound for the matched spectrum.** For the rest of this section assume  $\beta_0 = \beta_1 = \beta$ . Then  $P_w = \lambda_w^\beta$ ,  $\chi_R = \beta h$ , and

$$t_\delta = (\delta/M)^{1/\beta}, \quad b_\delta = -\log t_\delta.$$

Lemma 6.1 uses the tree stopped at  $\lambda_w \leq t_\delta$  or at depth  $N = n - J$ . Put  $c_i = -\log a_i$ ,  $c_{\min} = \min_i c_i > 0$ ,  $c_{\max} = \max_i c_i$ , and  $\Gamma_{n,\delta} = \beta(b_\delta + c_{\max})/n$ . A threshold leaf has logarithmic angular cost in  $[b_\delta, b_\delta + c_{\max}]$ ; an unfinished horizon leaf has cost below  $b_\delta$ . Thus every nonempty leaf of length  $\ell \leq n$ , containing  $j$  zeros with  $p = j/\ell$ , satisfies

$$(6.2) \quad \ell \chi(p) < b_\delta + c_{\max}, \quad \frac{\ell}{n} \leq \min\{1, \Gamma_{n,\delta}/[\beta \chi(p)]\}.$$

The corresponding term of the binomial law of parameter  $j/\ell$  gives  $\binom{\ell}{j} \leq e^{\ell H_b(j/\ell)}$ . End-point types have one word. Each leaf type therefore has at most  $e^{nS(\Gamma_{n,\delta})}$  words. There are at most  $(n+1)^2$  length/type pairs, including a possible empty word. By the actual catalogue construction,

$$(6.3) \quad r_F(n, \delta; Q) \leq 2^J (n+1)^2 e^{nS(\Gamma_{n,\delta})}.$$

No nonempty fixed-radius slice or full terminal image is assumed.

Every function under the maximum defining  $S$  is Lipschitz in  $\gamma$ , with common constant at most  $\log 2/(\beta c_{\min})$ . Their maximum is continuous. Here  $\beta b_{\delta_n} = -\log \delta_n + \log M$ , so  $\Gamma_{n,\delta_n} \rightarrow \gamma$  for every precision sequence. The fixed transient and polynomial factors have zero rate, giving

$$(6.4) \quad \limsup_n n^{-1} \log r_F(n, \delta_n; Q) \leq S(\gamma).$$

This includes  $\gamma = 0$  and nonmonotone sequences.

The reference angular leaf weights sum to one, and each is at least  $a_* t_\delta$ : a threshold leaf has parent weight above  $t_\delta$ , whereas an unfinished leaf has weight above  $t_\delta$  itself. This identity concerns a reference tree, not a physical partition. Enlarging a constant for  $n < J$  yields the useful coarse bound

$$(6.5) \quad r_F(n, \delta; Q) \leq C_U \delta^{-1/\beta} \quad (0 < \delta \leq 1), \quad r_F(n, \delta; Q) \leq 2^n.$$

**6.2. Shorter physical prefixes give the lower bound.** Let  $\gamma > 0$  and fix  $p \in (0, 1)$ . Choose  $j_L \in \{1, \dots, L-1\}$  with  $j_L/L \rightarrow p$ , and put

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L).$$

The exact identity and elementary type bounds

$$\binom{L-2}{j-1} = \frac{j(L-j)}{L(L-1)} \binom{L}{j}, \quad \frac{e^{LH_b(j/L)}}{L+1} \leq \binom{L}{j} \leq e^{LH_b(j/L)}$$

give  $B_L \rightarrow H_b(p)$  and  $\chi_L \rightarrow \chi(p)$ . For the lower type bound the  $j$ -th term is a largest term of the binomial law with parameter  $j/L$ , so is at least  $1/(L+1)$ . The additional fraction has logarithm  $o(L)$  for interior  $p$ .

The radial block cost in Lemma 5.2 is now  $c_L = \beta\chi_L$ . Fix  $L$  first and take

$$(6.6) \quad 0 < t < \min\{1, \gamma/(\beta\chi_L)\}.$$

The lemma, which checks the physical margin and every constrained time, gives

$$(6.7) \quad \liminf_n n^{-1} \log r_F(n, \delta_n; Q) \geq tB_L.$$

For fixed  $L$  take the supremum over the strict  $t$ , next let  $L \rightarrow \infty$ , and finally take the supremum over  $p \in (0, 1)$ . Endpoint frequencies have zero entropy, so continuity yields  $\liminf n^{-1} \log r_F(n, \delta_n; Q) \geq S(\gamma)$ . No witness radius or margin is required to remain uniformly positive as  $L \rightarrow \infty$ ; they are fixed in each horizon limit. For  $\gamma = 0$ , the bound  $r_F \geq 1$  and (6.4) give the zero limit. This proves (2.15).

**6.3. Positive-area and fixed near-full initial sets.** To complete Theorem 2.3, apply Theorem 2.5 to a fixed positive-area compact  $K$  and its positive-area set  $T$ . A catalogue covering  $K$  has at least  $\text{area}(T)/\sup_u \text{area}(T \cap A_u(n, \delta_n; K))$  elements. For fixed  $e \in (0, h)$ , put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + e} \right\rfloor \right\}.$$

Eventually  $m_n \geq 0$ , and  $\delta_n \leq e^{-(\chi_R + e)m_n}$ . Each term on the right of (2.21) is then at most a constant times  $e^{-(h-e)m_n}$ . Consequently

$$\liminf_n n^{-1} \log r_F(n, \delta_n; K) \geq (h - e) \min\{1, \gamma/(\chi_R + e)\}.$$

Since  $\chi_R = \beta h$  here, letting  $e \downarrow 0$  gives the lower bound in (2.16); for  $\gamma = 0$  the bound  $r_F \geq 1$  also suffices. The upper bound follows from  $r_F(n, \delta; K) \leq r_F(n, \delta; Q)$  and (2.15). The evaluation of  $S$  below will give the claimed common low-side rate.

For  $\zeta \in (0, 1)$  take the single near-full set  $T$  constructed in Section 4 and put  $K_\zeta = T \cup \{0\}$ . All exactly feasible length- $n$  words meeting  $T$  have, after their  $J$ -digit transient, frequency uniformly tending to  $a_0$ . If  $d_N \rightarrow 0$  bounds this deviation for tails of length  $N = n - J$ , there are at most

$$2^J(N+1) \exp \left( N \max_{|p-a_0| \leq d_N} H_b(p) \right) = e^{(h+o(1))n}$$

such words. Every state of  $T$  has an exactly feasible prefix, so these words, extended by any infinite tail, cover  $T$  through time  $n$ . One additional input covers the origin. This gives an upper rate  $h$  for every precision sequence, without selecting a strategy. Bound (6.5) also gives upper rate  $\gamma/\beta$ . The lower bound already proved gives (2.17). The construction of  $T$  used all-prefix frequency convergence, removal of null critical pullbacks, and compact inner approximation; it preceded every exponent, precision sequence and horizon. This completes Theorem 2.3 subject to the following elementary evaluation.

#### 6.4. Evaluation of the maximum.

*Proof of Corollary 2.4.* Reparameterise the maximum by the frequency  $q$  of the smaller branch  $\alpha = \alpha_*$ . This only changes the optimisation variable, not the dynamics or control assignments. Then  $\chi = c_b + d_b q$ . Write  $x = \gamma/\beta$ . The entropy inequality

$$c_b + d_b q - H_b(q) = q \log(q/\alpha) + (1 - q) \log((1 - q)/(1 - \alpha)) \geq 0$$

follows from  $-\log v \geq 1 - v$  and extends to the endpoints by continuity; equality holds at  $q = \alpha$ . For  $x \leq h$ , the objective  $H_b(q) \min\{1, x/(c_b + d_b q)\}$  is at most  $x$  and equals  $x$  at  $\alpha$ .

For  $h < x < c_b + d_b/2$  put  $q_x = (x - c_b)/d_b \in (\alpha, 1/2)$ . When  $q \leq q_x$  the objective is  $H_b(q) \leq H_b(q_x)$ . When  $q \geq q_x$  it is  $xH_b(q)/(c_b + d_b q)$ . This ratio decreases for  $q > \alpha$ : the numerator of its derivative is

$$D(q) = (c_b + d_b q)H'_b(q) - d_b H_b(q), \quad D(\alpha) = 0, \quad D'(q) = (c_b + d_b q)H''_b(q) < 0.$$

The maximum is therefore  $H_b(q_x)$ . Finally, for  $x \geq c_b + d_b/2$ , the value at  $q = 1/2$  is  $\log 2$ , the universal entropy maximum. Since  $\alpha < 1/2$ ,  $c_b + d_b/2 > c_b + d_b \alpha = h$ . The three pieces match at the thresholds, proving (2.18)–(2.19) and the low-side assertion of Theorem 2.3. Subtracting (2.17) gives (2.20). Strict increase of  $H_b$  on  $(\alpha, 1/2)$  proves the stated strict gap and the exact saturation threshold.  $\square$

#### 7. PHYSICAL PROBABILITY AND EXPONENTIAL RELIABILITY

Throughout this section the powers match,  $r_i = a_i^\beta$ , and (2.11) holds. We first prove probability domination on the whole triangle. The lower bound then uses regions with known positive physical mass, rather than only the individual witnesses needed for the full-state precision spectrum.

**7.1. All feasible transient branches.** Let  $J$  and  $\rho$  be as in Section 4, so every exactly feasible  $J$ -prefix ends in  $Q_\rho$ . Recall  $D_w$ , the complete exactly feasible domain, and  $w^-$ , the word with its last digit removed.

**Lemma 7.1** (Whole-triangle probability domination). *There is a fixed  $K_J < \infty$  such that for every  $1 \leq |w| \leq J$  and measurable  $B_0 \subset \mathbb{R}^2$ ,*

$$(7.1) \quad \mu(D_{w^-} \cap F_w^{-1} B_0) \leq K_J \text{area}(B_0).$$

Consequently, for every family  $\mathcal{V}$  of words of a common length  $N$ ,

$$(7.2) \quad \mu\left(\bigcup_{\substack{|\sigma|=J \\ v \in \mathcal{V}}} D_{\sigma v}\right) \leq C_0 \sum_{v \in \mathcal{V}} \lambda_v, \quad C_0 = \max\{1, 2^J K_J C_A \rho^2\}.$$

The constants are independent of  $N$ , the word family, tolerance, failure budget, and horizon.

*Proof.* Compactness gives  $d_0 = \min_{i,x \in Q} |\det DF_i(x)| > 0$ . On  $D_{w^-}$  all inputs to the derivative factors of  $F_w$  belong to  $Q$ , even when its last output leaves  $Q$ . Thus  $|\det DF_w| \geq d_0^{|w|}$  there. Cover this compact domain by finitely many local inverse charts for  $F_w$ , with compact closures inside regular charts. Change of variables bounds the pulled-back area in each chart by its inverse-determinant supremum times  $\text{area}(B_0)$ . Sum over the charts and maximise over the finitely many words of length at most  $J$ . All inverse branches meeting the domain are included; their overlap only enlarges the upper bound. No global inverse is used.

An exactly feasible  $\sigma$  sends  $D_\sigma$  into  $Q_\rho$ . Feasibility of the tail  $v$  places the endpoint in its actual complete region  $\mathcal{I}_v$ , whose area is at most  $C_A \rho^2 \lambda_v$  by (3.7). A possible endpoint at zero has a null pullback by (7.1). Pull back each region and sum over all at most  $2^J$  transients and all  $v$ . This proves (7.2).  $\square$

This is domination, not equality of word probabilities or a claim that feasible regions form a partition. Applying (7.1) to the first true exit during the transient would bound its area by  $JK_J C_Q \delta$ . We do not discard this area at failure cost  $e_n$ :  $\delta_n$  can be exponentially larger than  $e_n$ . The next construction uses exactly feasible prefixes.

**7.2. Two catalogue upper bounds.** Fix  $\varepsilon > 0$ ,  $n > J$ , and  $N = n - J$ . Keep every exactly feasible length- $n$  prefix  $\sigma v$ , with  $|\sigma| = J$ , whose tail zero frequency  $p$  satisfies  $I_a(p) \leq \alpha + \varepsilon$ . Empty complete regions may be omitted. Append any infinite input to each kept horizon prefix. Every state not covered has an exactly feasible length- $n$  prefix by controlled invariance, and all such prefixes from that state have discarded tails. By Lemma 7.1, its total probability is at most

$$(7.3) \quad C_0 \sum_{I_a(j/N) > \alpha + \varepsilon} \binom{N}{j} e^{-N\chi(j/N)} \leq C_0(N+1)e^{-N(\alpha+\varepsilon)} < e_n$$

for large  $n$ . Here  $\binom{N}{j} \leq e^{NH_b(j/N)}$  follows from the  $j$ -th term of the binomial probability sum. The catalogue has at most

$$(7.4) \quad 2^J(N+1)e^{NL_a(\alpha+\varepsilon)}$$

members. The fixed transient and polynomial factors have zero rate. Continuity of  $L_a$ , proved below, gives upper rate  $L_a(\alpha)$ .

Independently, the full-state stopping construction in Lemma 6.1 gives upper rate  $S(\gamma)$ . It assigns an actual feasible prefix to each state; after a threshold leaf its common tail satisfies every remaining constraint. We need not combine the two catalogues: choose the cheaper one. Therefore

$$(7.5) \quad \limsup_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \leq \min\{S(\gamma), L_a(\alpha)\}.$$

**7.3. A fixed positive mass gives the low-side lower bound.** Take one regular compact  $T \subset Q$  of area greater than  $3/4$  from Theorem 2.5. For fixed  $0 < \varepsilon < h$ , put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\beta h + \varepsilon} \right\rfloor \right\}.$$

Then  $\delta_n \leq e^{-(\beta h + \varepsilon)m_n}$  and, using  $\chi_R = \beta h$ , (2.21) bounds the area covered inside  $T$  by one input by a fixed multiple of  $e^{-(h-\varepsilon)m_n}$ . A successful catalogue covers more than  $1/2$  area inside  $T$  eventually. The union bound and  $\varepsilon \downarrow 0$  give

$$(7.6) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \min\{h, \gamma/\beta\}.$$

Since  $S(\gamma) = \gamma/\beta$  for  $\gamma \leq \beta h$  and  $L_a(\alpha) \geq h$ , this matches (7.5) on the entire low side. At  $\gamma = 0$ , the upper bound and  $r_{e_n} \geq 1$  give the zero limit. Fixed loss is used here to retain a positive successful mass, not to construct an upper bound with exponentially small failure.

**7.4. Positive-area regions forcing complete prefixes.**

**Lemma 7.2** (Thickened physical words). *For every integer  $L \geq 2$  there are  $\tau_L, E_L > 0$  such that any family  $\mathcal{W}$  of distinct words of a common length  $m$ , each admitting an infinite extension with runs at most  $L$ , has pairwise disjoint measurable regions  $E_w \subset Q$  with*

$$(7.7) \quad \mu(E_w) = \tau_L \lambda_w.$$

*Every point of  $E_w$  is exactly feasible under  $w$ . If  $n \geq m$  and  $2\delta < E_L P_w$ , every input whose actual successful set  $A_u(n, \delta; Q)$  meets  $E_w$  has prefix  $w$ . All constants are fixed before  $m, n, \delta$  and the word family.*

*Proof.* For each word choose a bounded-run extension  $v(w)$ . Lemma 5.1 gives a common radius bound  $s_L > 0$ , continuous centres  $y_w(s)$  for  $0 < s \leq s_L$ , and wrong-control excess  $d'_L > 0$ . Its construction gives centre angles at distance at least  $\Delta_L = 3d_L/4$  from both boundaries at every step, where  $d_L = 2a_*^{L+1}$ . Put  $K_0 = 1/a_* + 3B\rho$  and choose

$$0 < \epsilon_L \leq \min\{1/(2C_A), \Delta_L/(4C_A), d'_L/(4K_0C_A)\}.$$

Define

$$(7.8) \quad E_w = \{(s, sy) : s_L/2 \leq s \leq s_L, |y - y_w(s)| \leq \epsilon_L \lambda_w\}.$$

Fix the initial radius. On the feasible component between the centre and any feasible angle, (3.6) bounds the change in the angle after each prefix  $w|j$  by

$$(7.9) \quad |Y_{w|j}(y) - Y_{w|j}(y_w(s))| \leq C_A \epsilon_L \lambda_w / \lambda_{w|j} \leq D_L, \quad D_L = C_A \epsilon_L.$$

If the complete interval  $I_w(s)$  ended within the proposed initial interval, continuity would make an initial or intermediate angular constraint attain an endpoint. Its centre is at distance  $\Delta_L$ , while  $D_L \leq \Delta_L/4$ , a contradiction. The radii stay positive and below  $\rho < 1$ , so no radial constraint terminates the interval. Thus the entire interval is feasible. This argument applies even when the complete terminal image is truncated; it does not assume that image is onto.

At each prefix the thickened state and its centre lie on one image graph with logarithmic radial slope at most one. Hence their log radius ratio has absolute value at most  $D_L \leq 1/2$ , and their radius difference is at most  $2\rho D_L$ . The alternative angular map changes by at most

$$(1/a_* + B\rho)D_L + B(2\rho D_L) = K_0 D_L \leq d'_L/4.$$

Every thickened point therefore retains a wrong-control excess at least  $d'_L/2$  at every prefix, as well as selected-control feasibility. Continuity of the centres makes  $E_w$  measurable. Its actual area is

$$2\epsilon_L \lambda_w \int_{s_L/2}^{s_L} s \, ds = \frac{3\epsilon_L s_L^2}{4} \lambda_w; \quad \tau_L = 3\epsilon_L s_L^2/4.$$

Two distinct equal-length words cannot share a point in their regions: at their first difference one selected control would be wrong, contradicting feasibility under both words.

If an arbitrary input first differs at  $j \leq m$ , the preceding trajectory is exactly the selected one. Its radius is at least  $C_R^{-1}(s_L/2)P_{w|j-1} \geq C_R^{-1}(s_L/2)P_w$ . The wrong output has radial factor at least  $r_*/2$  and angular excess at least  $d'_L/2$ . Thus

$$(7.10) \quad |z_j| - s_j \geq E_L P_w, \quad E_L = \frac{s_L r_* d'_L}{8C_R} > 0.$$

Either  $z - s$  or  $-z - s$  has this positive excess. Each function has Euclidean Lipschitz constant  $\sqrt{2}$  and is nonpositive on  $Q$ , so the distance is at least  $E_L P_w / \sqrt{2} > \delta$ . The constraint is violated at that time. Later reentry cannot remove the violation. This proves the all-point forcing assertion.  $\square$

**7.5. Rare physical types and completion of the proof.** Suppose  $c = \gamma/\beta > h$ . Fix an interior frequency  $p$  with

$$(7.11) \quad \chi(p) < c, \quad I_a(p) < \alpha.$$

Choose  $j_L/L \rightarrow p$  and use the length- $L$  blocks starting with zero, ending with one, and containing  $j_L$  zeros. Write

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L), \quad I_L = \chi_L - B_L.$$

The binomial bounds proved in Section 6 give  $B_L \rightarrow H_b(p)$ ,  $\chi_L \rightarrow \chi(p)$  and  $I_L \rightarrow I_a(p)$ . Fix  $L$  large enough that  $\beta\chi_L < \gamma$  and  $I_L < \alpha$ . Put  $k = \lfloor n/L \rfloor$  and  $m = Lk$ . The  $M_L^k$

concatenations, extended by a fixed allowed block, have runs at most  $L$ . Lemma 7.2 gives disjoint regions of equal mass and equal radial product,

$$\mu(E_w) = \tau_L e^{-\chi_L m}, \quad P_w = e^{-\beta \chi_L m}.$$

The physical margin exceeds  $2\delta_n$  eventually because its logarithmic ratio has rate  $\gamma - \beta \chi_L > 0$ . One arbitrary input can then cover points in at most one region. A catalogue of  $R$  inputs successful on  $1 - e_n$  area must obey

$$(7.12) \quad R \geq M_L^k - \frac{e_n}{\tau_L e^{-\chi_L m}} = M_L^k \left( 1 - \frac{e_n}{\tau_L e^{-I_L m}} \right).$$

The parenthesis tends to one, since  $\alpha > I_L$ . Take the horizon limit with  $L$  fixed, then let  $L \rightarrow \infty$ . This proves

$$(7.13) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \sup_{\chi(p) < c, I_a(p) < \alpha} H_b(p) \quad (c > h).$$

To evaluate this lower bound, let  $a = a_* < 1/2$  and use  $q$  for the frequency of the smaller branch. Define the relabelled functions

$$\hat{\chi}(q) = c_b + dq, \quad \hat{I}(q) = q \log(q/a) + (1 - q) \log((1 - q)/(1 - a)),$$

where  $c_b = -\log(1 - a)$  and  $d = \log((1 - a)/a)$ , with the endpoint convention  $0 \log 0 = 0$ . If  $a_0 = a$ , then  $q = p$ ; otherwise  $q = 1 - p$ . Thus  $\hat{\chi}(q) = \chi(p)$ ,  $\hat{I}(q) = I_a(p)$ , and  $H_b(q) = H_b(p)$ . This relabelling changes neither the dynamics nor either variational maximum. Put  $J_a = \hat{I}(1/2) > 0$ . Then  $h = \hat{\chi}(a) = H_b(a)$ , and Corollary 2.4 gives the three pieces of  $S$ . Strict convexity of relative entropy, together with monotonicity of  $H_b$  on  $[0, 1/2]$ , gives

$$(7.14) \quad L_a(\alpha) = \begin{cases} H_b(q_\alpha), & 0 < \alpha < J_a, \hat{I}(q_\alpha) = \alpha, a < q_\alpha < 1/2, \\ \log 2, & \alpha \geq J_a. \end{cases}$$

This also proves continuity, including at  $J_a$ . For  $c > h$  set  $q_c = \min\{1/2, (c - c_b)/d\}$ , and set  $q_\alpha = 1/2$  if  $\alpha \geq J_a$ . Every  $a < q < \min\{q_c, q_\alpha\}$  satisfies (7.11). Approaching its endpoint in (7.13) yields

$$\liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq H_b(\min\{q_c, q_\alpha\}) = \min\{S(\gamma), L_a(\alpha)\}.$$

Together with (7.5) and (7.6), this proves Theorem 2.1. The common-cost endpoint  $c = h$  uses the fixed-mass argument, while the saturation and reliability endpoints use strict types approaching from below. Integer rounding changes  $m$  by less than fixed  $L$ ; the transient changes  $N$  by fixed  $J$ . The strict gaps in (7.3) and (7.12) absorb every  $o(n)$  fluctuation in both sequences. No constants are required to be uniform as  $L \rightarrow \infty$ , or over different fixed systems.

**Corollary 7.3** (Linear and fixed-failure specialisations). *The matched linear pair has rate (2.12) for every stated pair of sequences. Under the weaker standing null-critical hypotheses with matched powers, each fixed  $e \in (0, 1)$  instead satisfies*

$$(7.15) \quad \lim_n n^{-1} \log r_e(n, \delta_n; Q) = \min\{h, \gamma/\beta\}.$$

*Proof.* The linear pair is nonsingular on  $Q$ , so Theorem 2.1 applies. For fixed failure, choose the single compact set in Theorem 2.3 with area greater than  $1 - e$ ; its catalogue gives the upper rate in (7.15). For the lower rate choose a regular window  $T$  of area greater than  $(1 + e)/2$ . A catalogue with failure at most  $e$  covers at least  $\text{area}(T) - e > 0$  area there. The fixed-mass proof above, which needs only Theorem 2.5, gives the lower rate.  $\square$

## 8. A TRANSIENT PRESERVING FULL COSTS BUT CHANGING RELIABILITY

We prove Theorem 2.2. The construction alters only the first step from  $Q$ . It remains injective and has no positive-area collapse. Its small target regions have physical preimages whose probabilities decay too slowly to omit at the required failure budget.

**8.1. A flat increasing Cantor interpolation.** Fix the parameters of the theorem and put  $\lambda = a_0 a_1$ . The inverse reference maps are  $f_0(y) = a_0 y - a_1$  and  $f_1(y) = a_1 y + a_0$ . For the control blocks 01 and 10, write

$$B_0 = f_0 \circ f_1 = \lambda y + a_0^2 - a_1, \quad B_1 = f_1 \circ f_0 = \lambda y + a_0 - a_1^2.$$

Their invariant interval has endpoints and length

$$(8.1) \quad A = \frac{a_0^2 - a_1}{1 - \lambda}, \quad B = \frac{a_0 - a_1^2}{1 - \lambda}, \quad \ell = B - A = \frac{2\lambda}{1 - \lambda}.$$

Indeed  $A + 1 = 2a_0^2/(1 - \lambda) > 0$  and  $1 - B = 2a_1^2/(1 - \lambda) > 0$ , so  $[A, B] \subset (-1, 1)$ . The maps take this interval to its left and right portions of relative length  $\lambda < 1/2$ . The corresponding Cantor set  $\mathcal{C}$  consists of angles with infinite block controls 01, 10.

On  $J_0 = [-1/2, 1/2]$  use the source maps

$$b = 2/5, \quad h_0(y) = by - (1 - b)/2, \quad h_1(y) = by + (1 - b)/2.$$

Their Cantor set  $\mathcal{D}$  is null, because the total level- $k$  interval length is  $(2b)^k \rightarrow 0$ . For  $\sigma \in \{0, 1\}^k$  put

$$J_\sigma = h_{\sigma_1} \circ \cdots \circ h_{\sigma_k}(J_0), \quad Z_\sigma = B_{\sigma_1} \circ \cdots \circ B_{\sigma_k}([A, B]).$$

Within each family these closed intervals are separated by positive gaps. Their lengths are  $b^k$  and  $\ell\lambda^k$ , respectively. Both address maps preserve order and are unique, including endpoints. The target Cantor set is null as  $(2\lambda)^k \rightarrow 0$ .

We now construct a strictly increasing  $C^2$  homeomorphism  $g : \mathbb{R} \rightarrow \mathbb{R}$  satisfying

$$(8.2) \quad g(y) = y \quad (|y| \geq 1), \quad g([-1, 1]) = [-1, 1], \quad g(J_\sigma) = Z_\sigma, \quad g'(y) = 0 \iff y \in \mathcal{D}.$$

Let  $E(t) = e^{-1/t}$  for  $t > 0$  and zero otherwise, and set

$$\phi(t) = E(t)E(1 - t), \quad \Sigma(t) = \frac{\int_0^t \phi(v) dv}{\int_0^1 \phi(v) dv} \quad (0 \leq t \leq 1),$$

extended by zero and one outside that interval. The function is smooth, strictly increasing on  $(0, 1)$ , and flat at both endpoints. Map each point of  $\mathcal{D}$  to the point of  $\mathcal{C}$  with the same address. On a source gap  $(u, v)$  whose corresponding target gap is  $(U, V)$ , define

$$(8.3) \quad g(y) = U + (V - U)\Sigma((y - u)/(v - u)).$$

The ordered endpoint correspondence and increasing interpolation give the full interval identity  $g(J_\sigma) = Z_\sigma$ . At level  $k$ , the gap lengths are  $(1 - 2b)b^k$  and  $(1 - 2\lambda)\ell\lambda^k$ . For  $j = 1, 2$  the gap derivatives obey

$$(8.4) \quad |g^{(j)}| \leq C_j(\lambda/b^j)^k.$$

Since  $\lambda < 2^{-10} < b^2$ , both bounds tend to zero with level. Define  $q_1, q_2$  by these derivatives inside gaps and by zero on  $\mathcal{D}$ . They are continuous, by (8.4) for high-level gaps and endpoint flatness for the remaining finite collection.

These candidate derivatives are actual derivatives. The integral of  $q_1$  over a gap is its target gap length. Nullity of  $\mathcal{C}$  and order preservation therefore imply  $g(x) = A + \int_{-1/2}^x q_1(y) dy$  on  $J_0$ ; inside a gap include the partial integral of (8.3). Moreover

$$\sum_{\text{gaps}} \int |q_2| \leq C \sum_{k \geq 0} (2\lambda/b)^k < \infty.$$

Each complete-gap integral of  $q_2$  is zero, and a partial-gap integral equals  $q_1(x)$ . Since  $\mathcal{D}$  is null,  $q_1(x) = \int_{-1/2}^x q_2(y) dy$ . The fundamental theorem of calculus gives  $g' = q_1$  and  $g'' = q_2$ . Thus  $g$  is  $C^2$  on  $J_0$ , flat to second order on  $\mathcal{D}$ , with positive first derivative on all gaps.

For completeness, the exterior joins can be made without altering these properties. On  $[L, R]$  with prescribed increment  $T > 0$ , set  $d = \min\{(R - L)/4, T/4\}$  and  $v(x) = E(x - L)E(R - x)$ . For the left join use  $c(x) = 1 - \Sigma((x - L)/d)$ , and for the right use  $c(x) = \Sigma((x - R + d)/d)$ . In either case  $\int c \leq d < T$ . Integrate the positive interior density

$$(8.5) \quad q(x) = c(x) + \frac{T - \int_L^R c}{\int_L^R v} v(x).$$

For  $(-1, -1/2, A + 1)$  start at  $g(-1) = -1$ ; for  $(1/2, 1, 1 - B)$  start at  $g(1/2) = B$ . The identity endpoint has derivatives 1, 0, and the Cantor endpoint has derivatives 0, 0. Extending by identity beyond  $[-1, 1]$  therefore gives a global  $C^2$  map satisfying (8.2). It is strictly increasing and onto, and  $|g(y) - y| \leq 2$  globally. The map is fixed independently of every horizon and both sequences.

**8.2. The actual maps and all dynamical assumptions.** Set

$$\psi(s) = \frac{E(s - 1/4)}{E(s - 1/4) + E(1/2 - s)}.$$

Define

$$H(s, z) = \begin{cases} (s, z), & s \leq 1/4, \\ (s, z + s\psi(s)[g(z/s) - z/s]), & s > 1/4, \end{cases}$$

and set  $F_i = A_i \circ H$ . The smooth function  $\psi$  is zero on  $s \leq 1/4$  and one on  $s \geq 1/2$ . The quotient is used only away from zero; its needed derivatives are locally bounded at every joining point, and flatness of  $\psi$  gives global  $C^2$  maps. Near the origin  $H$  is identity, so the required first derivatives agree exactly.

On every fibre the angular map  $g_s(y) = (1 - \psi(s))y + \psi(s)g(y)$  is an increasing homeomorphism of  $[-1, 1]$  onto itself. Therefore  $H(Q) = Q$ , and one of the two linear controls is feasible at  $H(x)$  for each  $x \in Q$ . Its radial factor is less than one. This proves controlled invariance including every boundary and the origin.

Put  $m_* = \max_i \{a_i^\beta, a_i^{\beta-1}\} < 1$ ,  $\Lambda = 1 + 10m_*/(1 - m_*)$ , and  $V = \max\{\Lambda|s|, |z|\}$ . The bound  $|H^z - z| \leq 2|s|$  gives, on the whole plane,

$$V(Hx) \leq (1 + 2/\Lambda)V(x), \quad V(A_i x) \leq m_*(1 + 1/\Lambda)V(x).$$

Consequently the common contraction factor can be taken as

$$\theta = m_*(1 + 1/\Lambda)(1 + 2/\Lambda) \leq m_*[1 + 5/\Lambda] < (1 + m_*)/2 < 1.$$

This applies to arbitrary input tails outside  $Q$  as well.

For  $s > 1/4$  the determinant is

$$\det DH = 1 - \psi(s) + \psi(s)g'(z/s), \quad \det DF_i = a_i^{2\beta-1} \det DH;$$

for  $s \leq 1/4$  the first determinant equals one. Thus the critical set in  $Q$  is exactly  $\{(s, sy) : 1/2 \leq s \leq 1, y \in \mathcal{D}\}$ . It has area zero by Fubini in physical coordinates. Local inverse charts off that null set show that every null target set has a null pullback in  $Q$ . No positive-area set is collapsed to a null set. Each fibre map is strictly increasing and identity outside  $[-1, 1]$ , so  $H$  is globally bijective. It is proper, since its first coordinate is unchanged and  $|H^z - z| \leq 2|s|$ . Thus it and both  $F_i$  are global homeomorphisms. They are not asserted to be diffeomorphisms; the displayed critical points are genuine.

The fixed radial factors satisfy

$$a_1^\beta = (1 - 2^{-10})^{4096} < e^{-4} < 1/4, \quad a_0^\beta < 1/4.$$

After the first input from  $Q$ , the radius lies below  $1/4$ , regardless of whether its angle is feasible. Every subsequent input therefore uses the identity part of  $H$ , even along an approximate trajectory that has left  $Q$ . For every input and every  $j \geq 1$ ,

$$(8.6) \quad F_u^j x = A_u^j Hx \quad (x \in Q).$$

At time zero both  $x$  and  $Hx$  belong to  $Q$ . It follows that  $A_u^F(n, \delta; Q) = H^{-1}A_u^A(n, \delta; Q)$  on  $Q$ . Since  $H(Q) = Q$ , full covers correspond, proving (2.13). Probability covers correspond only after transporting  $\mu$  to  $H_*\mu$ , which is absolutely continuous but is not uniform area. This is consistent with measure-transport invariance results; it is not a counterexample to them.

**8.3. Positive-area forced families and the strict rate gap.** Every infinite sequence of blocks 01, 10 has runs at most two. The reference endpoint-distance identities used in (5.1) give  $|y_j| \leq 1 - d$  at all shifted positions, with  $d = 2a_*^3$ . For a chosen zero, the current angle is at most  $a_0 - a_1 - a_0d$ , so the alternative one gives at most  $-1 - a_0d/a_1$ ; the symmetric bound holds for a chosen one. Thus every wrong-control angular excess is at least  $d_* = (a_*/a^*)d > 0$ . This holds on each complete interval  $Z_\sigma$  through its  $2k$ -digit block prefix: its endpoints have such bounded-run extensions, and the linear prefix maps are monotone. No symbolic Cantor subset replaces the whole interval in this assertion.

For each  $\sigma \in \{0, 1\}^k$  define an actual physical region

$$E_{\sigma,k} = \{(s, sy) : 1/2 \leq s \leq 1, y \in J_\sigma\}.$$

Here  $\psi = 1$  and  $g(J_\sigma) = Z_\sigma$ . The regions are disjoint and each has mass  $(3/8)b^k$ , so their union has mass  $(3/8)(2b)^k$ . Every point requires its selected  $2k$ -digit control prefix at sufficiently small tolerance. To check this for an arbitrary input, let its first differing digit be  $j \leq 2k$ . Equation (8.6) and  $s \geq 1/2$  give the preceding physical radius at least  $\frac{1}{2}\lambda^{\beta k}$ : a partial block product before  $2k$  is no smaller than the complete product  $\lambda^k$ . The wrong output radial factor is at least  $a_*^\beta$  and its angular excess is at least  $d_*$ . Therefore

$$(8.7) \quad |z_j| - s_j \geq c_0 \lambda^{\beta k}, \quad c_0 = \frac{1}{2} a_*^\beta d_* > 0.$$

Its distance from  $Q$  is at least this value divided by  $\sqrt{2}$ . For  $k = \lfloor n/2 \rfloor$  and  $\gamma = -\beta \log \lambda$ , the logarithmic ratio to  $\delta_n$  has positive rate  $-\frac{1}{2}\beta \log \lambda$ . Every successful input thus has the full selected prefix, for every point in the region. Later reentry cannot remove the constraint at this first wrong step.

One catalogue input can consequently cover points of at most one region. A catalogue of  $R$  inputs with failure at most  $e_n$  obeys

$$(8.8) \quad R \geq 2^k \left( 1 - \frac{8e_n}{3(2b)^k} \right).$$

Since  $-\frac{1}{2} \log(2b) = \frac{1}{2} \log(5/4) < 1/8 = \alpha$  by  $\log(1+x) < x$ , the parenthesis tends to one. Thus the lower limit is at least  $\frac{1}{2} \log 2$ . Both strict exponent gaps absorb all the allowed  $o(n)$  fluctuations.

Finally  $S(\gamma) = \log 2$ : at  $p = 1/2$  the threshold ratio  $\gamma/[\beta\chi(p)]$  is two. For  $q = 1/16$ ,

$$H_b(q) = \frac{1}{4} \log 2 + \frac{15}{16} \log(16/15) < \frac{1}{4} \log 2 + \frac{1}{16} < \frac{1}{2} \log 2,$$

and

$$I_a(q) > \frac{5}{8} \log 2 - H_b(q) > \frac{3}{8} \log 2 - \frac{1}{16} > \frac{1}{8},$$

using  $\log 2 > 1/2$ . Relative entropy is strictly increasing for  $q > a_0$ ; its sublevel  $I_a(q) \leq 1/8$  therefore has right endpoint below  $1/16$ . Binary entropy is increasing up to  $1/2$ , so  $L_a(1/8) < H_b(1/16) < \frac{1}{2} \log 2$ . This proves the strict inequality (2.14) and completes Theorem 2.2.

The example leaves the full-state finite-horizon counts unchanged but changes the probability weights of forced physical regions. Its own accurate reliability spectrum and the

existence of its reliability limit are not determined here; the strict lower-limit gap suffices for the obstruction. Absolute continuity and null-set pullback preservation alone do not control exponential tail masses.

### 9. A NONLINEAR EXAMPLE WITH INWARD BOUNDARY IMAGES

In this section the powers match:  $\beta_0 = \beta_1 = \beta > 1$ , so  $r_i = a_i^\beta$  and  $t_i = a_i^{\beta-1}$ . The following nonsingular example has nonlinear, truncated feasible images. It satisfies the reliability hypotheses and cannot be reduced to the matched linear pair by the stated constraint-preserving simultaneous conjugacies.

**9.1. Empty fixed-radius slices.** Let  $t = \max a_i^{\beta-1} < 1$ ,  $\Lambda = 1 + 4t/(1 - t)$ ,  $v = t(1 + 1/\Lambda) < 1$ , and fix

$$0 < c < \min\{1 - a^*, (1 - v)/v, 1\}, \quad g(s) = 1 - \frac{cs}{1 + s^2}.$$

Consider the single explicit family

$$(9.1) \quad \begin{aligned} F_0(s, z) &= (r_0 s, g(s)t_0(z + a_1 s)), \\ F_1(s, z) &= (r_1 s, g(s)t_1(z - a_0 s)). \end{aligned}$$

Both are global smooth diffeomorphisms:  $g \in [1 - c/2, 1 + c/2]$  is positive, the first coordinate has the explicit inverse  $s = s'/r_i$ , and the second is an affine invertible function of  $z$  at that radius. Their first jets are (2.4). For the stated  $V$ ,

$$(9.2) \quad V(F_i x) \leq \max\{r^*, (1 + c/2)v\} V(x), \quad \max\{r^*, (1 + c/2)v\} < 1.$$

Thus the norm inequality is checked globally, not inferred from feasible trajectories.

For  $0 < s \leq 1$ ,  $a^* < g(s) < 1$ . The complete one-step angular domains are

$$(9.3) \quad D_0(s) = [-1, a_0/g(s) - a_1], \quad D_1(s) = [a_0 - a_1/g(s), 1].$$

Their endpoints lie strictly inside  $(-1, 1)$ , and their overlap has angular width  $1/g(s) - 1 > 0$ , physical width  $s(1/g(s) - 1) = O(cs^2)$  near zero. This proves controlled invariance and the actual overlap geometry. Theorem 2.3 applies.

Yet for every fixed  $0 < s_0 \leq 1$ , a sufficiently long all-zero word is infeasible for every initial angle. To verify this, write  $s_k = r_0^k s_0$  and  $q_k = y_k + 1$ . Under zero input,

$$(9.4) \quad q_{k+1} = \frac{g(s_k)}{a_0} q_k + 1 - g(s_k).$$

The multiplier is uniformly greater than one, since  $g(s) > a^*$ , and  $q_1 \geq 1 - g(s_0) > 0$  for  $y_0 \geq -1$ . The sequence eventually exceeds two, violating  $y \leq 1$ . Monotonicity means that this also holds for the smallest initial angle, hence for all of them. Thus a description in which every word has a nonempty full-image slice does not apply to this example.

This statement concerns  $I_0^m(s_0)$  at a fixed radius, not the whole physical region  $\mathcal{I}_0^m$ . Choosing the radius smaller can make a given finite word feasible. Indeed any finite word admits a bounded-run infinite extension, and Lemma 5.1 realises that extension at a sufficiently small positive radius. Missing fixed-radius slices must not be confused with removing a word everywhere in  $Q$ .

There is also a constraint-preserving conjugacy obstruction. For (9.1),  $F_0$  sends the positive lower edge of  $Q$  strictly inside  $Q$ , and its upper edge outside;  $F_1$  does the symmetric thing. The image of the top edge has constant radius  $r_i < 1$  and crosses the opposite angular boundary exactly once. Thus

$$(9.5) \quad F_i(\partial Q) \cap \partial Q \text{ consists of exactly two points.}$$

In contrast, for any target pair whose zero control sends the whole lower edge of  $Q$  into  $\partial Q$  and whose one control sends the whole upper edge into  $\partial Q$ , both corresponding intersections contain a continuum. A homeomorphism  $H$  of the plane with  $H(Q) = Q$ ,

simultaneously conjugating the two labelled controls (even allowing a label permutation), would map each intersection in (9.5) onto the corresponding target intersection. Finite versus continuum is impossible. Thus this example is not reducible to such a boundary-edge-preserving pair by a fixed  $Q$ -preserving simultaneous control homeomorphism, and in particular not by such a  $C^1$  or bi-Lipschitz map. No assertion about unconstrained coordinate changes or every member of the new class is made.

For example, take  $a_0 = 1/3$ ,  $a_1 = 2/3$ ,  $\beta = 2$  and  $c = 1/100$ . The choice  $\Lambda = 9$  in (9.2) gives a global factor at most  $67/90 < 1$ . Corollary 2.4 evaluates this same inward family at exponent  $\gamma = \log 4$ :

$$2 \left( \log 3 - \frac{2}{3} \log 2 \right) < \log 4 < \log(9/2), \quad \lim_n n^{-1} \log r_F(n, \delta_n; Q) = H_b \left( \frac{\log(4/3)}{\log 2} \right).$$

This holds for every precision sequence of that exponent. The first inequality follows from  $(4/3)^3 > 2$  and the second from  $4 < 9/2$ . It is an evaluation of the spectrum, not a claim that missing fixed-radius words require a different count.

## 10. CONSEQUENCES AND SCOPE

The same contracting system has several distinct information costs. A fixed tolerance gives zero asymptotic full-cover cost. At exponentially shrinking tolerance, matched first derivatives determine the complete spectrum  $S$ . One fixed nearly full-area compact set has the smaller rate  $\min\{h, \gamma/\beta\}$ , while an exponentially reliable catalogue must also account for rare physical regions. With no critical points on  $Q$ , its exact rate is  $\min\{S(\gamma), L_a(\alpha)\}$ .

The two geometric ingredients have different uses. The first-true-exit area estimate applies to each arbitrary approximate input and survives finite transients after a fixed small area loss. The whole-triangle transport and positive-area forcing estimates preserve the probabilities needed for exponential reliability. The Cantor transient separates these properties: every full-state finite-horizon count is unchanged, but reliability has a strictly larger cost. Nonsingularity on  $Q$  is sufficient for the transfer theorem, and null critical sets alone are insufficient for a universal transfer. Neither statement is a necessary and sufficient classification.

The results retain binary planar controls, the specified first-order matrices, a triangular controlled invariant constraint, and common norm contraction toward the origin. The norm condition is not contraction between arbitrary pairs of states. The probability is uniform area on the same physical triangle; a transported measure defines a different comparison. The successful sets in a reliability catalogue may vary with the horizon, whereas the nearly full-area compact set is chosen before all precision exponents and sequences. The mismatch criterion is limited to the stated matrix family. No high-precision mismatch spectrum, classification of all compact initial sets, fixed-strategy optimality, or exact reliability spectrum of the counterexample is asserted.

## APPENDIX A. MISMATCH AND THE COMMON-COST CRITERION

**Theorem A.1** (First-order matching criterion). *For every fixed system satisfying the standing assumptions, the following property holds if and only if  $\beta_0 = \beta_1$ : there are  $\gamma_1 > 0$  and a function  $c : (0, \gamma_1) \rightarrow [0, \infty)$  such that, for every fixed positive-area compact  $K \subset Q$ , every  $\gamma \in (0, \gamma_1)$ , and every precision sequence of exponent  $\gamma$ ,*

$$\lim_n n^{-1} \log r_F(n, \delta_n; K) = c(\gamma).$$

**Theorem A.2** (Low-precision mismatch). *Suppose  $\beta_0 \neq \beta_1$ . For each  $\zeta \in (0, 1)$  there is one compact  $K_\zeta \subset Q$ , of area greater than  $1 - \zeta$ , such that simultaneously for every  $0 < \gamma < \gamma_0$  and every precision sequence of that exponent,*

$$(A.1) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = \gamma D, \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \gamma h / \chi_R < \gamma D.$$

The set is selected before the exponent, sequence, and horizon. Every other fixed positive-area compact  $K \subset Q$  has lower rate at least  $\gamma h / \chi_R$  in this range.

We now return to the standing assumptions with  $\beta_0 \neq \beta_1$ . The same physical estimates apply, but the angular and radial products no longer have a common-power identity.

*Proof of Theorem A.2.* First consider the upper rate on  $Q$ . For every  $0 < \gamma < \gamma_0$ , all branches of the tree in Lemma 6.1 cross  $P_w \leq \delta_n/M$  before depth  $n - J$  for large  $n$ , since

$$P_w \leq e^{-\gamma_0(n-J)} < \delta_n/M \quad \text{at that depth.}$$

The ratio has logarithmic rate  $\gamma - \gamma_0 < 0$ , so the statement includes every  $o(n)$  tolerance fluctuation. For the uncapped radial tree, write  $N(t)$  for its leaf count. Replacing a parent by its children preserves  $\sum P_w^D = 1$ , and each leaf satisfies  $r_* t < P_w \leq t$ . Hence

$$(A.2) \quad t^{-D} \leq N(t) \leq (r_* t)^{-D}, \quad r_F(n, \delta_n; Q) \leq 2^J N(\delta_n/M).$$

This proves the upper rate  $\gamma D$  through an actual full-time catalogue, without asserting that the reference tree is necessary.

For the lower rate put  $p_* = r_0^D$ , so  $1 - p_* = r_1^D$ , and  $c_R(p) = -p \log r_0 - (1 - p) \log r_1$ . Then

$$(A.3) \quad H_b(p_*) = D c_R(p_*).$$

Choose  $j_L/L \rightarrow p_*$  and use the blocks in Lemma 5.2, with  $c_L = c_R(j_L/L)$  and  $B_L = L^{-1} \log \binom{L-2}{j_L-1}$ . The binomial calculation in Section 6 gives  $B_L \rightarrow H_b(p_*)$  and  $c_L \rightarrow c_R(p_*)$ . For each fixed  $L$ ,  $c_L \geq \gamma_0 > \gamma$ . Every strict  $0 < t < \gamma/c_L < 1$  therefore gives lower rate  $t B_L$ . Take its supremum first, then let  $L \rightarrow \infty$ . Equation (A.3) gives the lower rate  $\gamma D$ . The positive-radius witnesses and wrong-control margins are those of the common physical lemma, not midpoints of assumed full images. This proves the first limit in (A.1).

For any fixed positive-area compact  $K$ , take the set  $T$  in Theorem 2.5, and set

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + e} \right\rfloor \right\}.$$

For large  $n$  it is nonnegative, and  $\delta_n \leq e^{-(\chi_R + e)m_n}$ . Every term of (2.21) is then bounded by a constant times  $e^{-(h-e)m_n}$ . Covering the positive area of  $T$  implies lower rate  $\gamma(h - e)/(\chi_R + e)$ , since  $\gamma < \gamma_0 \leq \chi_R$ . Let  $e \downarrow 0$ . This proves the asserted universal lower bound  $\gamma h / \chi_R$ .

For  $K = Q$ , choose the single set  $T$  of area greater than  $1 - \zeta$  in the area proof, and put  $K_\zeta = T \cup \{0\}$ . Keep only stopped pairs in Lemma 6.1 realised on  $T$ . They cover it with the complete tails already constructed. Let  $q_\delta = -\log(\delta/M)$  and  $c_{\max} = -\log r_*$ . Each retained tail of length  $\ell$  has

$$q_\delta \leq -\log P_w < q_\delta + c_{\max}, \quad \ell \leq \frac{q_\delta + c_{\max} + \log D_e}{\chi_R - e},$$

by (4.6). At this length,  $\lambda_w \geq D_e^{-1} e^{-(h+e)\ell}$ , while all reference length- $\ell$  weights sum to one. There are at most  $D_e e^{(h+e)\ell}$  such words. Sum over these lengths and the at most  $2^J$  transients, and include one input for the origin. For fixed  $e \in (0, h)$  this gives

$$(A.4) \quad \log r_F(n, \delta_n; K_\zeta) \leq \frac{h + e}{\chi_R - e} [q_\delta + c_{\max} + \log D_e] + O_e(\log(1 + q_\delta)) + O_e(1).$$

Because  $q_\delta = \gamma n + o(n)$ , letting  $e \downarrow 0$  gives upper rate  $\gamma h / \chi_R$ , matching the lower bound. The set was selected through uniform frequency convergence before  $e$ , the precision exponent, the sequence, and the horizon.

Finally let  $b_i = r_i^D$ . The classical relative-entropy identity is

$$D\chi_R - h = \sum_i a_i \log(a_i/b_i) \geq 0.$$

Concavity of the logarithm proves the inequality, with equality only when  $b_i = a_i$  for both labels. Equality would require  $\beta_0 D = \beta_1 D = 1$ , contrary to mismatch. Thus  $D > h/\chi_R$ , completing the theorem.  $\square$

*Proof of Theorem A.1.* If  $\beta_0 = \beta_1 = \beta$ , Theorem 2.3 gives the common rate  $c(\gamma) = \gamma/\beta$  on  $(0, \beta h)$ . If  $\beta_0 \neq \beta_1$ , Theorem A.2 separates  $Q$  and one fixed nearly full-area compact set for every  $0 < \gamma < \gamma_0$ . Any interval adjacent to zero contains such an exponent, contradicting the common-rate property.  $\square$

The criterion concerns precisely the first derivatives (2.4), nonsingularity almost everywhere in  $Q$ , controlled feasibility, and the common norm inequality. The proof does not extend this necessity statement to arbitrary feedback systems, classify high-precision mismatch rates, or compare a fixed strategy with the optimal catalogue.

## APPENDIX B. FINITE-HORIZON COUNTS AND THE CONTROL ALPHABET

This finite-horizon example separates equality of exponential rates from equality of catalogue counts when the available controls change. Let  $q \geq 2$  be an integer,  $0 < \rho < 1$ , and consider

$$f_v(s, z) = (\rho s, \rho(qz + sv)), \quad U_q = \{q - 1 - 2j : 0 \leq j < q\}, \quad \widehat{U}_q = [-(q - 1), q - 1].$$

Use the same triangle  $Q$ , and put  $K_1 = [1/2, 3/4] \times [-1/4, 1/4] \subset \text{int } Q$ . Here  $r_U^{\text{ex}}(n; K)$  denotes the minimum number of input words that keep every assigned state of  $K$  in  $Q$  exactly at times  $0, \dots, n$ , using alphabet  $U$ . This illustrative family is separate from the unequal-branch standing assumptions.

For either alphabet,

$$r_U^{\text{ex}}(n; Q) = q^n, \quad q^n/2 \leq r_U^{\text{ex}}(n; K_1) \leq q^n.$$

Indeed, at positive radius the angular map is  $y \mapsto qy + v$ . For the digit alphabet the inverse one-step intervals tile  $[-1, 1]$ , and their compositions produce  $q^n$  exactly feasible intervals of length  $2q^{-n}$ , including every intermediate constraint. They give the upper bounds for both alphabets. For any fixed word over the larger alphabet, the final angular map has slope  $q^n$ . Its feasible initial angles therefore have length at most  $2q^{-n}$ . Applying this bound to all of  $[-1, 1]$ , or to the slice  $s = 1/2$  of  $K_1$  with angles  $[-1/2, 1/2]$ , gives the two lower bounds. Consequently both exact exponential rates are  $\log q$  for both alphabets.

The minimum count on  $K_1$  can nevertheless decrease. For  $q = 2$  and  $n = 1$ , input 1 fails on the positive-angle end of  $K_1$ , and input  $-1$  fails on its negative-angle end. Two digit controls are necessary and sufficient. Over  $[-1, 1]$ , input 0 sends every angle of  $K_1$  to  $2y \in [-1, 1]$  and keeps the actual radial coordinate in  $(0, 1)$ . Thus

$$r_{\{-1, 1\}}^{\text{ex}}(1; K_1) = 2, \quad r_{[-1, 1]}^{\text{ex}}(1; K_1) = 1.$$

Enlarging the control set preserves the displayed full-triangle counts and rate bounds, not all finite-horizon counts.

## DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

Generative AI, including ChatGPT (OpenAI), was used extensively in the research and preparation of this draft. Its uses included developing mathematical arguments and examples, checking proof chains and source comparisons, and drafting, restructuring and compiling the English LaTeX manuscript. The research arguments are given in full in the text; AI-assisted checks do not constitute independent verification by a human author. Human authorship, affiliations, the extent of human mathematical review, and approval of the manuscript and this disclosure have not yet been confirmed. These facts must be completed by the responsible human authors before submission.

## REFERENCES

- [1] H. Chen, Y. Huang and X. Zhong, Variational principles of invariance entropy dimension, *J. Differential Equations*, **453**, Part 1 (2026), 113819. doi:10.1016/j.jde.2025.113819.
- [2] Z. Chen and X. Zhong, Invariance complexity and equi-invariability for control systems, *J. Math. Anal. Appl.*, **539** (2024), 128533. doi:10.1016/j.jmaa.2024.128533.
- [3] F. Colonius, Invariance entropy, quasi-stationary measures and control sets, *Discrete Contin. Dyn. Syst.*, **38** (2018), 2093–2123. doi:10.3934/dcds.2018086.
- [4] F. Colonius and C. Kawan, Invariance entropy for control systems, *SIAM J. Control Optim.* **48** (2009), no. 3, 1701–1721. doi:10.1137/080713902.
- [5] I. Csiszár, The method of types, *IEEE Trans. Inform. Theory*, **44** (1998), 2505–2523. doi:10.1109/18.720546.
- [6] J. E. Hutchinson, Fractals and self-similarity, *Indiana Univ. Math. J.*, **30** (1981), 713–747. doi:10.1512/iumj.1981.30.30055.
- [7] C. Kawan, *Invariance Entropy for Deterministic Control Systems: An Introduction*, Lecture Notes in Mathematics, 2089, Springer, 2013. doi:10.1007/978-3-319-01288-9.
- [8] C. Kawan and A. Da Silva, Invariance entropy for a class of partially hyperbolic sets, *Math. Control Signals Syst.*, **30** (2018), Article 18. doi:10.1007/s00498-018-0224-2.
- [9] X. Nie, T. Wang and Y. Huang, Measure-theoretic invariance entropy and variational principles for control systems, *J. Differential Equations*, **321** (2022), 318–348. doi:10.1016/j.jde.2022.03.013.
- [10] T. Wang and Y. Huang, Inverse variational principles for control systems, *Nonlinearity*, **35** (2022), 1610–1633. doi:10.1088/1361-6544/ac4f33.
- [11] T. Wang, Y. Huang and H.-W. Sun, Measure-theoretic invariance entropy for control systems, *SIAM J. Control Optim.*, **57** (2019), 310–333. doi:10.1137/18M1197862.
- [12] R. Yang, E. Chen, J. Yang and X. Zhou, Bowen’s equations for invariance pressure of control systems, *SIAM J. Control Optim.*, **63** (2025), 1104–1128. doi:10.1137/23M1607684.